Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems

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***Abstract*—Clinical multimodal Internet-of-Things data cannot be centralized; hierarchical federated learning addresses that but supplies no privacy guarantee. Such systems guarantee privacy over records, while clinical practice requires it over individuals; the two differ wherever an episode of care yields many records. We present a three-tier framework whose privacy unit is the patient: gradients are grouped by individual before bounding, noise is calibrated against every client so the edge tier enters the budget, and the conformal quantile is charged to it. Across two deployments the composed budget is $\varepsilon \leq 1$, uplink falls by at least $90.8\,\%$, energy per round by at least $72.3\,\%$, and 95th-percentile inference latency is at most 3.0 ms; collapsing the edge tier does not improve accuracy, so the savings cost no utility. Pretrained initialization, not head-only noise, makes a strict budget survivable: tightening ε from ∞ to 0.5 costs a randomly initialized arm 0.118 in area under the ROC curve and a pretrained one nothing measurable; restricting noise to the head adds nothing at any budget. The price of the guarantee grows with the records each individual contributes, vanishing where the units coincide, which provides a built-in control; against a forecast frozen before an external corpus of 187 hospitals it reproduces in shape but not magnitude. Fusion is conditional on balance between streams: worth 0.0370 on one cohort, untestable on a second at ceiling, and null on a third until its streams are equalized. An individual-level membership-inference audit with a passing positive control refutes the predicted leakage difference without establishing equivalence.**

***Index Terms*—Hierarchical federated learning, Internet of Things, differential privacy, user-level privacy, membership inference, multimodal learning, edge computing, quality of service, conformal prediction, clinical decision support.**

I. Introduction

Internet-of-Things deployments increasingly gain their value from combining different kinds of sensing streams. An operating room records continuous physiological waveforms together with the structured clinical history of the patient. A wearable installation records inertial motion together with egocentric video. In both cases the streams complement each other, and a model that sees both can recognize states that neither reveals alone. The streams are also produced where the sensors are, which is rarely where computation is convenient. This work addresses the tension between where data arises and where models are trained.

The obvious solution is to send the raw streams to a central service, but this is often not possible. The data volume is large and the uplink is the scarce resource, so centralization increases both latency and energy in exactly the deployments that can least afford them. More importantly, in the settings where multimodal sensing is most valuable, the data cannot legally move at all. Clinical records are governed by consent and by data-protection laws that prohibit pooling across institutions. A model that requires centralization is therefore not merely expensive; it cannot be deployed. Federated learning [1] solves this by exchanging model updates instead of records. Hierarchical variants [2], [3] add an edge tier, so that the wide-area link is used only once every several local rounds.

Moving the model instead of the data does not, by itself, protect the data. The updates a client sends are computed from its records, and membership inference against released models is well established [4]. A formal guarantee is therefore needed. Differentially private stochastic gradient descent [5] provides one by bounding the influence of each training record and adding noise to the aggregate. However, the guarantee is stated over the unit that is clipped, and in a sensing deployment that unit is a fixed-length window, not a person. Every patient, and every participant in a continuous recording, contributes many windows. A record-level budget answers whether a particular window was used in training, but a clinical deployment must answer whether a particular *person* was. Group privacy [6] can relate the two. At the group sizes found in these datasets, however, the conversion is prohibitive: a nominal budget of one becomes a budget that supports no meaningful claim.

Existing systems have developed along two separate lines. Hierarchical designs make a convincing case for communication and energy savings. HED-FL [3] extends deployment lifetime by avoiding long-range transmission, HCEF [7] adapts computation and compression to device heterogeneity, and assignment-oriented work [8] reduces divergence under non-identical data distributions. None of them provides a formal privacy guarantee. Private designs do provide one: FL2DP [9] combines noise with shuffling, and RoPPFL [10] pairs local privacy with robust aggregation at the edge. Neither is hierarchical in the sense that matters here, and neither examines how an intermediate tier changes how often each client's budget is spent. In both lines the local objective is treated as the aggregation of generic parameter vectors. Neither the presence of several modalities per record nor the grouping of records by individual plays any role.

Three difficulties follow, and they affect deployability as

much as privacy. First, the unit at which a guarantee is stated is rarely examined, so its cost on genuinely paired multimodal data has not been quantified. Second, parameter-efficient private training is valid only if the corpus used to pretrain the frozen representation is disjoint from the protected population [11], [12]. This condition is often left implicit, and where it is respected, its cost is not measured. Third, guarantees are usually reported from the parameters of the mechanism rather than verified by attack. Where an audit is performed, it is done at the level of the record, which is exactly what a record-level guarantee already protects.

Two remedies follow directly. First, the unit can be fixed at the point of clipping instead of being recovered afterwards through group privacy. If the gradients of an individual's records are summed before the bound is applied, that individual's influence on the released model is bounded no matter how many records they contributed. Accounting at the same level then yields a budget that is, by construction, a statement about people. Second, the resulting guarantee can be audited by membership inference at the level of the individual. For a negative finding to be informative, the attack must be strong enough: it must be shown to find leakage on a configuration that is known to memorize.

Accordingly, this work presents PP-HFM, a three-tier federated framework for multimodal sensing in which the privacy unit is the individual by construction. Records are grouped by individual before they are bounded. The noise scale is calibrated to meet the budget for every client at once, not for a single representative client, so the reported figure reflects the effect of the edge tier. The cost of releasing a conformal calibration quantile is included in that figure rather than left out. Representation learning is separated from private adaptation: the encoders are pretrained on a population disjoint from every client and then frozen, so only a compact fusion head is privatized. The framework is evaluated on two corpora whose modalities are linked by the recording apparatus rather than by a pairing rule. The protocol measures communication, latency, reliability and energy on a simulated topology, and it audits the privacy guarantee by attack.

The paper makes five contributions.

First, the framework enforces an individual-level guarantee within a hierarchical multimodal system and reports a composed budget that covers both training and calibration. The accounting this requires is specified in enough detail to be checked by others: the noise is calibrated against every client rather than a representative one, and the calibration release is charged to the same budget. An error in either is not detectable from a training log.

Second, the cost of the choice of unit is measured on genuinely paired clinical multimodal data. So is the cost of the disjointness condition that makes the guarantee valid end to end. Neither is assumed to be negligible.

Third, the guarantee is audited by patient-level membership inference, using both a threshold attack and a likelihood-ratio attack [13]. Each attack is reported together with a positive control on a configuration known to memorize. The audit returns a null on the private arms, and the control attributes that null to the defense rather than to a weak adversary.

Fourth, the system-level behavior is validated on two deployments that differ in topology and payload. Every network quantity is measured on rounds in which the payload actually crossed the simulated topology, not computed analytically. The same protocol shows that the guarantees are affordable in real time. The conformal set construction, which carries the coverage guarantee, takes $0.68\,\%$ of the inference path. The private mechanism takes none of it, because its noise is added to gradients during training and adds no operation at inference.

Fifth, the price of the individual-level guarantee is shown to be not a fixed quantity but a function of how much each individual contributes. With the per-step budget and the number of units per step held constant, the penalty is indistinguishable from zero when each individual contributes a single record. It grows monotonically as that number rises and saturates by eight records. At one record per individual the two units coincide by construction, so this configuration serves as a control, and the divergence at larger contributions is attributable to the unit rather than to the experimental setup.

User-level differential privacy [14], parameter-efficient differentially private fine-tuning [11], [12] and differentially private quantile release [15] are established prior art and are used, not proposed, in this work. The contributions are measurements of these mechanisms in a setting where they have not previously been evaluated, together with the corresponding auditing practice.

The rest of the paper is organized as follows. Section II reviews related work on hierarchical and private federated learning, on the unit at which a guarantee is stated, and on membership inference as an auditing tool. Section III presents the proposed framework. Section IV describes the corpora, the implementation, the evaluation measures and the experimental protocol. Section V reports the results, Section VI discusses their interpretation and the limitations of the study, and Section VII concludes.

II. Related Work

A. Federated Learning and Hierarchical Aggregation

Federated learning trains a shared model by exchanging model updates instead of data [1]. In its flat form, every client communicates with a central server in every round, so the wide-area link, rather than local computation, limits how often the global model can be updated. Client–edge–cloud designs [2] address this with an intermediate tier that performs several rounds of local aggregation before each round of central aggregation. HED-FL [3] applies this principle to constrained deployments, clustering devices dynamically at the edge and extending the operational lifetime of an installation by avoiding long-range transmission. HCEF [7] adds heterogeneity awareness, adjusting per-device computation frequency and compression ratio online so that the slowest participant does not determine the round duration. Mhaisen *et al.* [8] optimize the assignment of clients to aggregators and show that an assignment based on the statistics of local data reduces model divergence under non-identically distributed data. These works establish the communication and energy benefits of hierarchy, which the present study adopts. None of them, however,

provides a formal privacy guarantee. HED-FL provides none, and the assignment strategy of [8] requires clients to disclose local label statistics, which is itself a release of information about the protected population.

B. Differential Privacy in Federated Learning

Differentially private stochastic gradient descent [5] bounds the influence of each training record by clipping its gradient to a fixed norm and adding calibrated noise to the aggregate. The accumulated privacy cost is tracked by a moments or Rényi accountant [16] within the framework of [6]. In federated learning, the mechanism has been developed in two main directions. FL2DP [9] combines an exponential noise mechanism with a shuffler that removes the association between a client and its update, improving accuracy at an equivalent guarantee. RoPPFL [10] combines local differential privacy with similarity-based robust aggregation at the edge, so that the intermediate tier also verifies updates. Both show that a formal guarantee is attainable in federated learning at acceptable cost. Neither, however, accounts for the edge tier: in FL2DP the edge plays no role in the privacy accounting, and neither work examines how an intermediate tier changes the frequency with which each client's budget is spent, which determines the calibration of the noise.

C. The Unit to Which a Guarantee Applies

The distinction between record-level and individual-level guarantees is well established. McMahan *et al.* [14] introduced user-level differential privacy for federated language models, bounding the contribution of each user rather than each example, and group privacy provides the standard conversion between the two [6]. What is missing is a measurement of the cost of the individual-level guarantee in the setting that most requires it. In clinical monitoring, every participant contributes many records, because a single episode of care yields many time-aligned windows. The conversion factor therefore applies to the entire population rather than to a few atypical participants. The literature reports neither the utility lost when the unit is moved from the record to the individual on genuinely paired multimodal clinical data, nor how that cost varies with the number of records per individual. Without the latter, a single measurement cannot be distinguished from a property of the particular cohort.

D. Parameter-Efficient Private Adaptation

The noise a parameter vector must absorb under a fixed budget grows with its dimension, so private training of a full network degrades severely at the budgets required for clinical use. Restricting private optimization to a compact trainable subset above frozen pretrained representations is an established remedy: Li *et al.* [11] and Yu *et al.* [12] show that parameter-efficient differentially private fine-tuning is considerably more robust at small ε than end-to-end private training. The present work adopts this approach and claims no novelty for it. The approach, however, is valid only if the corpus used to obtain the frozen representations is disjoint from the protected population. Otherwise, the representations are derived from that population without a guarantee, and the budget reported for the trainable component does not describe the system as a whole. This condition is often left implicit, and where it is satisfied, its cost in utility is rarely reported.

E. Membership Inference as Empirical Auditing

A differential privacy guarantee is an upper bound on the influence of a participant; it does not show what an adversary achieves in practice. Membership inference provides this complementary measurement. Shokri *et al.* [4] introduced the shadow-model formulation, and thresholding of the loss or confidence [17] remains the most commonly reported attack. Such attacks are weak, because a single global threshold ignores the varying difficulty of individual examples. An easily fitted non-member may therefore score above a poorly fitted member, and the reported attack strength is a lower bound on leakage rather than an estimate of it. Watson *et al.* [18] showed that calibrating for per-example difficulty changes conclusions substantially. Carlini *et al.* [13] formalized this calibration as a likelihood-ratio test between per-example in- and out-distributions estimated from shadow models, and recommended evaluating attacks by their true-positive rate at a low false-positive rate. Two limitations remain. First, membership is almost always evaluated at the level of the training record, which a record-level guarantee already protects, rather than at the level of the individual. Second, negative results are rarely accompanied by evidence that the attack could produce a positive one, so the absence of detected leakage cannot be distinguished from a weak adversary.

F. Multimodal and Clinical Federated Learning

Federated learning is particularly well motivated in medicine, where data governance rather than data volume prevents pooling at a single site [19], [20]. Most federated work, including the systems discussed above [3], [7], [9], [10], treats the local objective as the aggregation of generic parameter vectors and does not model multiple modalities per record, although clinical data are typically multimodal and the available modalities may differ between institutions. In existing multimodal federated benchmarks, the modalities are often paired by a construction rule rather than by common origin. An apparent fusion benefit on such a benchmark may reflect the pairing rule rather than complementary information, so it cannot support a claim about fusion. Corpora in which both modalities come from the same individual during the same episode and are aligned at capture [21], or synchronized in hardware [22], avoid this problem, because the pairing is a property of the source.

G. Uncertainty Quantification under a Privacy Budget

Conformal prediction produces prediction sets with a distribution-free coverage guarantee and requires neither retraining nor distributional assumptions [23], [24]. This makes it suitable for clinical deployment, where a point prediction without a confidence statement is of limited use. In a private

system, however, conformal calibration requires the release of a quantile of nonconformity scores computed on held-out data from the protected population, and this release is itself a disclosure. Differentially private quantile release by the exponential mechanism is standard [15]. The open issue is accounting: the cost of calibration composes with the cost of training, and a system that reports only the latter understates its total budget, by an amount that grows with the number of calibration releases.

H. Real-Time Inference and Quality of Service at the Edge

A federated model is trained once and queried continuously. After deployment, its clinical usefulness depends on the delay between the arrival of a measurement and the availability of a prediction, not on the time to convergence. The federated literature has studied the latter extensively and the former very little. HCEF [7] adapts computation frequency and compression ratio online, but it optimizes the duration of a training round. RoPPFL [10] verifies updates at the edge, prioritizing robustness over response time. HED-FL [3] minimizes the energy needed to train a model rather than the latency of using it. Inference cost has mainly been studied in the edge-computing literature rather than the federated literature [25], where reduced-precision arithmetic, in particular integer quantization, is an established way to lower memory footprint and inference time on commodity hardware [26]. A federated system may therefore meet a strict privacy budget and still be unusable at the bedside if its prediction latency exceeds the time within which a clinician can act. No existing federated system reports end-to-end inference latency under its deployed privacy mechanism together with the communication and energy costs of training the model.

I. Summary and Research Gap

Five gaps follow from this review. First, hierarchy and formal privacy have been developed largely separately: hierarchical systems provide no guarantee, and private systems do not examine how an intermediate tier affects budget expenditure. Their interaction is therefore unmeasured. Second, the mechanism for stating a guarantee over individuals is established, but its cost on genuinely paired clinical multimodal data is not. Without a characterization of how this cost varies with the number of records per individual, a single measurement cannot be distinguished from an artifact of the cohort. Third, parameter-efficient private adaptation is widely used, but the disjointness of the pretraining corpus on which its validity depends is often left implicit, and its cost is not reported. Fourth, guarantees are usually reported from the mechanism's parameters rather than audited by attack. Where audits exist, they operate at the record level, and null results are rarely accompanied by evidence that the attack could detect leakage. Fifth, the cost of federated designs is reported in training terms, such as rounds to convergence, bytes exchanged and energy consumed, while the inference latency experienced by the clinician is not measured, although it determines whether the system can be used at the point of care.

The present work addresses these five gaps together. It combines a two-tier hierarchy with a guarantee enforced at the level of the individual at the point of clipping, rather than obtained by conversion afterwards, and calibrates the noise against every client simultaneously so that the reported budget reflects the edge tier. It measures the utility lost by moving the unit from the record to the individual, and characterizes this cost as a function of the number of records per individual rather than at a single operating point. It enforces and verifies the disjointness of the pretraining population and reports the utility cost of an end-to-end valid guarantee. It audits the guarantee by individual-level membership inference, using both a threshold attack and a per-individual calibrated likelihood-ratio attack, each reported with a positive control on a configuration known to memorize. Finally, it reports the end-to-end prediction latency, the traffic carried by the hierarchy and the energy consumed by each tier for the same configuration that carries the privacy guarantee.

III. Methodology

A. Overall Framework of the Proposed PP-HFM

Multimodal Internet-of-Things deployments combine sensing streams of different kinds, such as physiological monitors with clinical records or inertial sensors with video, at the location where the streams are produced. The training data are therefore distributed, and in clinical settings they cannot be centralized, because consent and data-protection regulations prohibit pooling. Federated learning addresses this by moving the model instead of the data, but it does not provide a privacy guarantee, because client updates are functions of the client's records. A suitable framework must therefore combine distributed multimodal training with a formal guarantee, within the network, latency and energy constraints of an edge installation.

Differentially private federated learning provides such a guarantee, but the *unit* of the guarantee in clinical settings has received little attention. Standard differentially private stochastic gradient descent [5] clips and accounts per training record. In a monitoring dataset, a record is a fixed-length window rather than a person, and every patient in the cohort used here contributes several windows. A record-level budget therefore bounds whether a particular window was used in training, whereas a clinical guarantee must bound whether a particular *patient* was. Prior federated systems address only one side of this problem. HED-FL [3] provides a hierarchical tier without differential privacy, FL2DP [9] provides differential privacy without a hierarchy, and neither these nor the heterogeneity-aware and assignment-oriented designs [7], [8] model multiple modalities per record or the grouping of records by individual.

This work therefore proposes a framework in which the privacy unit is the patient by construction, and in which the hierarchy and the guarantee are designed jointly. The system, PP-HFM (Patient-Private Hierarchical Federated Multimodal learning), has three tiers: institutional clients that hold patient-grouped multimodal records and never release them, edge aggregators that perform several local rounds before communicating upward, and a cloud aggregator that maintains the

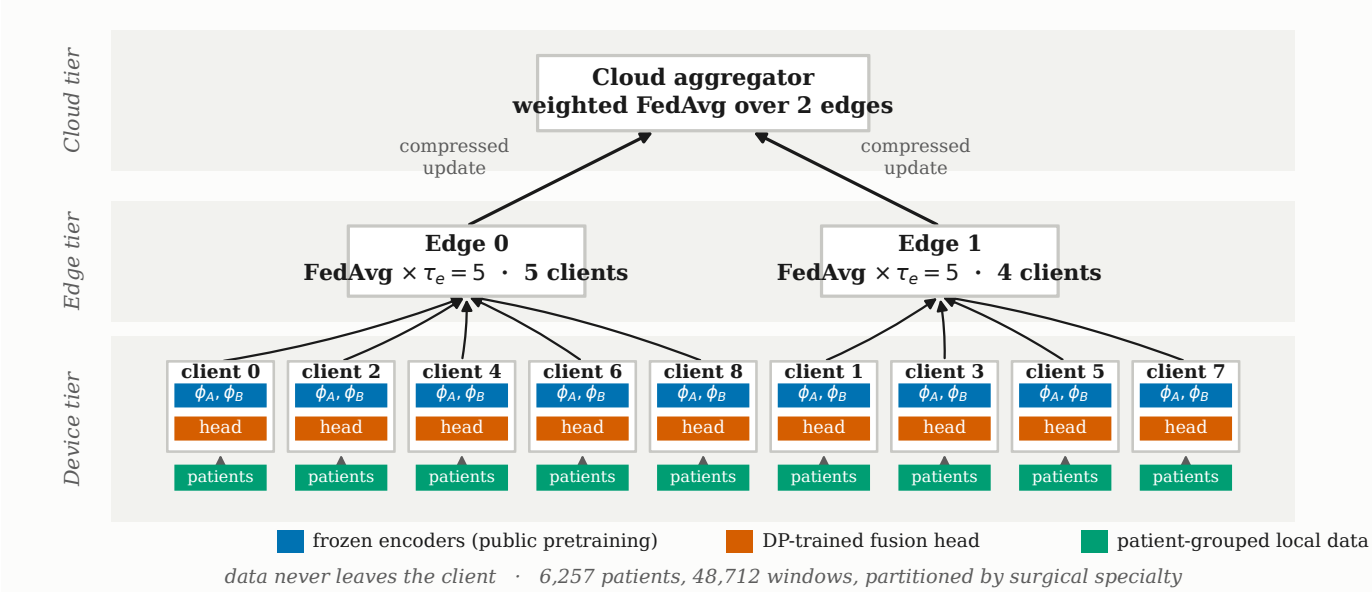


Fig. 1. Overall architecture of PP-HFM. Nine institutional clients, partitioned by surgical specialty, hold patient-grouped multimodal records that never leave the institution; each runs frozen pretrained encoders and privately trains only the fusion head. Two edge aggregators perform τ_e rounds of weighted averaging for every cloud round, and the cloud aggregator maintains and redistributes the global model. Clients are drawn grouped by edge for legibility; the indices give the true assignment.

global model. Within each client, representation learning is separated from private adaptation. Encoders pretrained on a disjoint public population are frozen, and only a compact fusion head is trained under a differentially private objective with patient-level sensitivity. Fig. 1 shows the overall organization.

Each client samples a batch of patients rather than windows, sums the gradients of all windows of each sampled patient into one per-patient gradient, clips that gradient, and adds noise to the aggregate once per optimization step. The influence of any individual on the released model is therefore bounded regardless of the number of records contributed. The resulting update is compressed and sent to the client's edge, which averages the updates of its clients. After several edge rounds, the edges report to the cloud, which averages across edges and redistributes the global model. At the end of training, the clients release differentially private calibration quantiles, from which a distribution-free prediction set is constructed, and the privacy cost of this release is charged to the same budget as training.

The framework comprises the following components. The patient-level privacy mechanism enforces sensitivity with respect to the individual. The multi-client noise calibration selects a single noise scale that satisfies the budget for every client despite large differences in client size. The two-tier aggregation produces the global model while using the wide-area link once per several local rounds. The update encoding reduces the transmitted payload. The federated conformal calibration provides coverage guarantees under the same budget. The deployment path specifies how the released model is executed within a latency budget, and the network and energy instrumentation measures the communication and energy costs of the hierarchy. An empirical audit tests the resulting guarantee. The encoder architecture, the aggregation rule and the conformal procedure are adopted from established work and are not claimed as contributions.

B. Overall System Architecture

PP-HFM is a three-tier federated system built on a frozen multimodal backbone. Responsibilities are separated: the encoders provide fixed representations, a compact head contains all trainable and privatized parameters, an edge tier absorbs local rounds to reduce wide-area traffic, and a cloud tier maintains the global model. The encoder design, the weighted averaging rule and the conformal calibration procedure are adopted from prior work. The design contribution lies in the unit at which privacy is enforced and in the accounting of the reported budget.

The architecture is shown in Fig. 1. An institutional client holds a set of patients, each contributing several time-aligned multimodal records. The records are encoded by two frozen modality-specific encoders and combined by a trainable fusion head. The head is optimized under a differentially private objective, and its compressed update is the only quantity that leaves the institution. Updates are averaged at the edge over the clients it serves. After τ_e such rounds, the edge models are averaged at the cloud and redistributed. No record, single-record gradient or uncompressed update crosses an institutional boundary.

The system consists of three functional groups. The first is the frozen backbone: two modality-specific encoders, pretrained on a population disjoint from every federated client and not adapted during federated training. The second is the privatized trainable component: the fusion head, the patient-level clipping and noise mechanism that governs its optimization, and the calibration procedure that sets the noise scale. The third comprises the communication and calibration com-

ponents: the update encoding, the two-tier averaging, and the federated conformal procedure that converts point predictions into prediction sets with a coverage guarantee.

Information flows in a group-then-bound structure. Within a client, records are encoded and combined by the head, and gradients are grouped *by patient* before clipping, so that the clipped quantity is a patient's total contribution rather than a single record's. Noise is added once to the aggregate of the clipped per-patient gradients, after which the update is compressed and transmitted. Communication is hierarchical: client to edge in every edge round and edge to cloud once per cloud round, with the ratio set by τ_e.

This design follows from the constraints of the setting. Grouping gradients by patient before clipping makes the guarantee a statement about individuals, and avoids the severe degradation of a group-privacy conversion. Restricting private optimization to a small head over frozen features keeps the dimension of the perturbed vector low, which is necessary for acceptable utility under a tight budget. Pretraining the features on a disjoint population keeps the guarantee valid end to end. The edge tier reduces wide-area transmissions by a factor of τ_e without changing the learning objective. Because the encoders are fixed and the aggregation rule is standard, observed differences can be attributed to the privacy unit and its accounting rather than to the representation or the averaging.

C. Frozen Multimodal Backbone

A tight privacy budget requires a large noise scale, and the noise a parameter vector must absorb grows with its dimension, so private training of a full network degrades severely. Parameter-efficient differentially private fine-tuning, which restricts private optimization to a small trainable subset over frozen pretrained features, is considerably more robust at small ε than end-to-end private training [11], [12]. PP-HFM adopts this design unchanged. The framework adds the condition under which the resulting guarantee remains valid, and a measurement of the cost of that condition.

The backbone consists of two modality-specific encoders. The temporal encoder takes a multichannel physiological time series as input. A multi-scale convolutional stem applies three parallel one-dimensional convolutions with different kernel widths to capture fine, intermediate and coarse temporal patterns, and concatenates their outputs. A strided convolution then reduces the temporal resolution, two residual convolutional blocks and a bidirectional recurrent layer model longer-range dependencies, and an attention-pooling layer aggregates over time with learned weights. Because pooling is adaptive, the encoder accepts sequences of any length. The static encoder processes a fixed-length vector of preoperative clinical and laboratory variables with a two-layer perceptron with normalization and a nonlinearity. The two embeddings are concatenated and mapped to class logits by the fusion head, as shown in Fig. 2.

Two properties of the backbone are required by the privacy mechanism. First, its parameters remain frozen throughout federated training, so all trainable and privatized capacity resides in the fusion head, which holds $559{,}874$ of the $1{,}864{,}227$ parameters. The private optimization is therefore close to a linear probe over fixed features. Second, every normalization layer is a group or layer normalization; no batch normalization is used. Batch normalization mixes statistics across the examples of a batch, so per-example, and therefore per-patient, gradients would not be well defined, and the privacy accounting would be invalid. The recurrent layer is likewise implemented in a form compatible with per-sample gradients.

The pretraining corpus must satisfy an additional condition. If the encoders are pretrained on data from the federated clients, they have been exposed to the protected population without a guarantee, and the budget reported for the head does not describe the system end to end. PP-HFM therefore distinguishes two encoder configurations.

In the *oracle* configuration, the encoders are pretrained on the pooled training data of the federated clients. This configuration does *not* provide a valid end-to-end guarantee and is not proposed for deployment. It is used to characterize the mechanism, because it isolates the attainable performance of the head-only recipe from the origin of the representation. In the *disjoint* configuration, a set of individuals is reserved before federated training, disjoint at the individual level from every client's training, validation and test splits, and the encoders are pretrained only on this set. Disjointness is verified explicitly. This configuration provides the guarantee claimed in this paper, and its utility cost is measured. Section IV-B specifies which configuration produced each result, and Section V-A reports the cost of moving between them.

D. Patient-Level Privacy Mechanism

A clinical system must provide a guarantee about individuals, whereas the standard mechanism provides one about records. Differentially private stochastic gradient descent bounds the influence of a training *record* by clipping its gradient to a fixed norm and adding noise to the aggregate [5]. When each record corresponds to a distinct person, the two coincide. In a monitoring dataset, however, a record is a fixed-length window and a single operation yields many windows, so removing one patient removes as many records as that patient contributed. An (ε, δ)-differentially private mechanism is $(k\varepsilon,\ ke^{(k-1)\varepsilon}\delta)$-differentially private for groups of size k [6]. Applied at the mean group size of this cohort, this conversion turns a nominal $\varepsilon = 1$ into $\varepsilon \approx 8$ with $\delta \approx 9 \times 10^{-2}$, which does not support a clinical claim. The gap between the two units is illustrated in Fig. 3.

Whether the two units differ depends on the deployment. When each client corresponds to a single individual, as in a wearable installation where one participant carries one device, client-level, user-level and individual-level guarantees coincide. When a client holds records from many individuals, as an institution holding many patients does, the units differ for every participant. The mechanism below addresses the second case. The framework applies to both, but only the second exhibits the gap, and the evaluation is organized accordingly.

The mechanism enforces the guarantee at the point of clipping rather than by conversion afterwards. The sampling

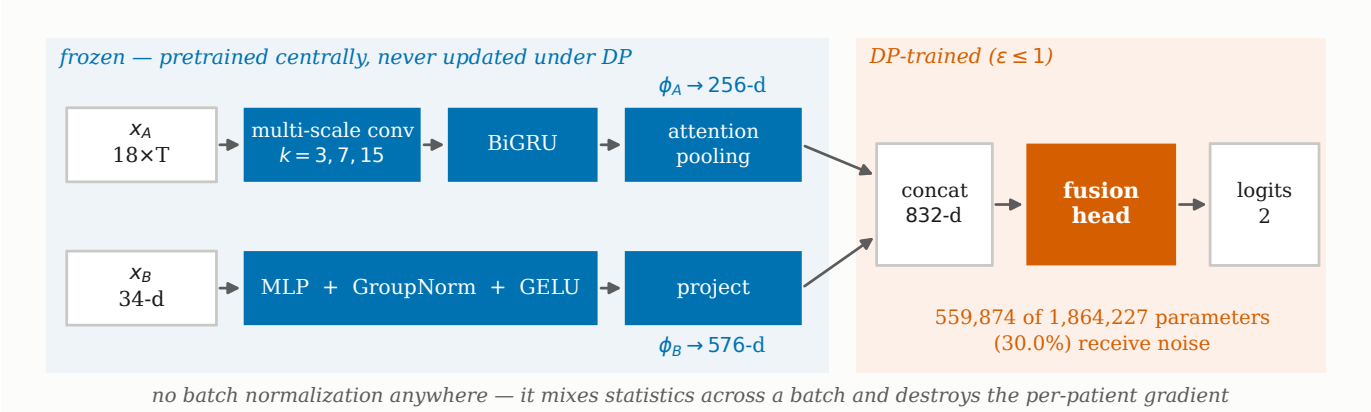


Fig. 2. Encoder–fusion backbone and the privacy boundary. Both encoders are pretrained centrally and frozen; only the fusion head is trained under differential privacy, so noise is applied to 559,874 of 1,864,227 parameters. Section III-C defines the two pretraining configurations, and Section IV-B records which produced which result. No batch normalization appears anywhere, because it mixes statistics across a batch and would make the per-patient gradient ill-defined.

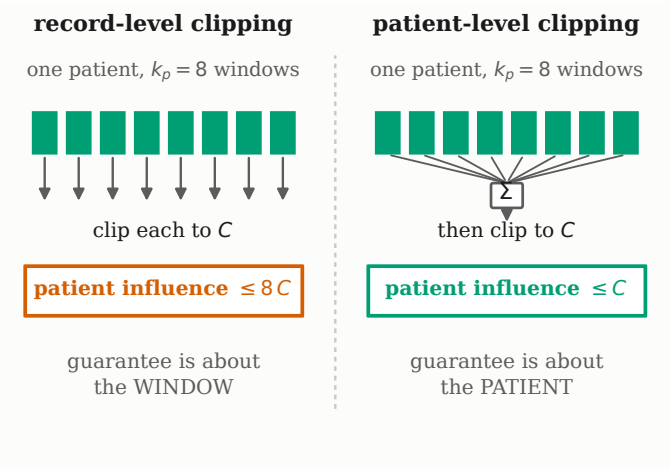


Fig. 3. Where the guarantee is decided. Clipping each window to C bounds the influence of a *window*: a patient contributing k_p windows can still displace the aggregate by k_pC. Summing a patient's windows before clipping bounds the influence of the *patient* at C, however many windows that patient contributed. k_p=8 is the cohort mean.

Algorithm 1 Patient-level private local update at client k

Require: frozen θ_{enc}; head θ_{head}; patients $\mathcal{P}_k$ with records $\{W_p\}$; clip C; noise σ; rate q from (3); step size η

for each local step **do**
  Sample $S \subseteq \mathcal{P}_k$, each p independently w.p. q
  for $p \in S$ **do**
    $g_p \leftarrow \sum_{(x,y)\in W_p} \nabla_{\theta_{\text{head}}} \ell(f_\theta(x), y)$ ▷ group by patient
    $\bar{g}_p \leftarrow g_p \cdot \min(1, C/\|g_p\|_2)$ ▷ bound the sum
  end for
  $\tilde{g} \leftarrow \frac{1}{|S|}\left(\sum_{p\in S} \bar{g}_p + \mathcal{N}(0, \sigma^2 C^2 I)\right)$
  $\theta_{\text{head}} \leftarrow \theta_{\text{head}} - \eta\, \tilde{g}$
  Accountant.step(σ, q)
end for
return θ_{head}

unit is the patient: at each optimization step, a client samples patients independently at a fixed rate, so the rate entering the accountant is defined over patients rather than windows. For each sampled patient, the gradients of all of that patient's windows are summed into one per-patient gradient, and this sum is clipped to the clipping norm. Clipping the sum makes the sensitivity patient-level: a patient can change the aggregate by at most the clipping norm regardless of the number of windows, whereas clipping each window separately would allow a change proportional to that number. Gaussian noise calibrated to the same norm is then added once per step to the aggregate of the clipped per-patient gradients, and the head parameters are updated. The mechanism is shown in Fig. 4 and stated in Algorithm 1.

Accounting uses the same patient-level sampling rate, so the reported budget is patient-level by construction. A single accountant instance persists for the whole of a client's participation, so composition across rounds is preserved. The construction has two costs, which this work measures: the number of effective units per step decreases from windows to patients, and a patient with many windows has their summed gradient clipped more strongly than a patient with few. The dependence of this cost on the number of records per individual is examined empirically by holding the first effect fixed and varying the second (Section V-B3).

E. Noise Calibration Across Heterogeneous Clients

The noise multiplier determines the budget for a given clipping norm, and depends on the sampling rate and on the number of optimization steps of a client. In a federated system partitioned by institution or specialty, these quantities differ substantially between clients: a small client samples a large fraction of its population per step but performs few steps, whereas a large client does the reverse. Neither case dominates in general. Because the reported budget is the maximum over clients rather than the average, a noise scale calibrated for any single client may be insufficient for another, and the reported guarantee would then be violated.

The calibration therefore selects the smallest noise multiplier that satisfies the target budget for every client simultaneously, by evaluating each candidate value against all clients.

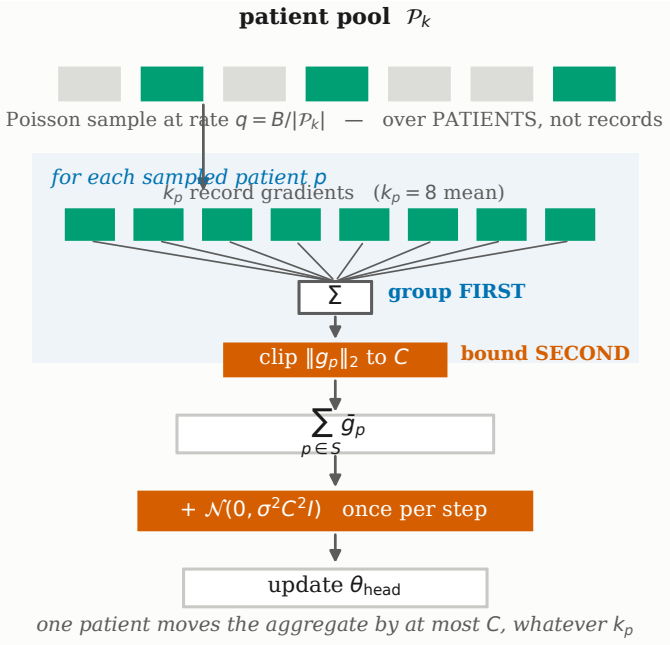


Fig. 4. Internal processing of the patient-level privacy mechanism. Patients are sampled, each sampled patient's window gradients are accumulated into one per-patient gradient, that sum is clipped to C, the clipped sums are aggregated, and a single Gaussian perturbation scaled to C is applied per step. Grouping before bounding is what fixes the unit of the guarantee.

Two details are essential. First, the number of optimization steps per call is computed by rounding the ratio of the client's population to the batch size *upward*; rounding downward undercounts the steps and therefore underestimates the required noise. Second, the population and sampling rate of each client are read from the partition files used in training rather than from cached metadata, because metadata that no longer matches the data produces a budget that is violated without any warning. Both details are stated as part of the method because violations of either are not detectable from training logs and can only be found by independent recomputation.

F. Two-Tier Hierarchical Aggregation

In a flat federated system, every client update is sent to the cloud in every round, so the wide-area link limits the update frequency. An intermediate aggregation tier removes this limitation [2], [3]: several rounds of local optimization and edge averaging are performed for each round of cloud averaging, so the wide-area link is used once per group of local rounds. PP-HFM adopts this organization, with τ_e edge rounds per cloud round, and claims no novelty for the averaging rule.

Aggregation is performed in two stages, each weighted by the number of training records of the participants. Within a cloud round, each client initializes from its edge model, performs one private local pass, and sends its encoded update. The edge computes the weighted average over its clients and returns it, and this is repeated τ_e times. The edges then report to the cloud, which computes the weighted average over edges and broadcasts it as the global model for the next round. Non-floating-point parameters are passed through unchanged. The procedure is given in Algorithm 2.

Algorithm 2 PP-HFM cloud round r

Require: global model $\theta^{(r-1)}$; edge map $e(\cdot)$; edge rounds τ_e

$\theta_e^{(r,0)} \leftarrow \theta^{(r-1)}$ for all edges e
for $s = 1$ **to** τ_e **do**
  for each client k **in parallel do**
    $\theta_k^{(r,s)} \leftarrow$ Algorithm 1 from $\theta_{e(k)}^{(r,s-1)}$
    Transmit $Q\big(\text{top-}\kappa(\theta_k^{(r,s)} - \theta_{e(k)}^{(r,s-1)})\big)$
  end for
  Each edge aggregates by (8)
end for
Cloud aggregates by (9) and broadcasts $\theta^{(r)}$
if final round **then**
  Clients release perturbed quantiles of (12); aggregate to $\hat{q}$
end if
return $\theta^{(r)}$, $\hat{q}$

The parameter τ_e controls the trade-off of the hierarchy. With $\tau_e = 1$, the edge tier is merged into the cloud tier, which recovers conventional single-tier federated averaging; this configuration is used to isolate the contribution of the hierarchy. Larger values reduce wide-area transmissions proportionally but allow edge models to diverge further between synchronizations. Because the privacy mechanism operates entirely within a client and aggregation uses only privatized updates, τ_e affects communication and convergence, and enters the guarantee only through the number of optimization steps, which the calibration already accounts for.

G. Update Encoding and Conformal Calibration

Two further components are adopted from established practice and are not claimed as contributions. The first reduces the transmitted payload. Each client update is sparsified by retaining only its largest-magnitude coordinates, and the retained values are uniformly quantized to a reduced bit width. The residual removed by sparsification is stored locally and added to the next round's update, so suppressed coordinates are transmitted later rather than lost. The encoding is applied after the privacy mechanism and is therefore post-processing; by the post-processing property of differential privacy, it does not affect the guarantee and can be tuned for bandwidth alone.

The second component provides uncertainty quantification. Conformal prediction produces prediction sets with a distribution-free coverage guarantee without retraining [23], [24]. Each client computes nonconformity scores on a held-out calibration split and releases a quantile of them; the edges aggregate the client quantiles, the cloud aggregates the edge quantiles, and the resulting global threshold defines the prediction set. The released quantile is a statistic of protected data, so it is perturbed before release and charged to the budget: the calibration term receives a fixed share of the total budget in advance, and the system reports the composed total. If calibration were performed once per edge round rather than once per training run, its cost would compose across all edge rounds, and reporting only the training term would understate

the total budget by a multiplicative factor. Allocating the budget in advance and reporting the composed total prevents this.

H. Deployment Path and Inference Under a Latency Budget

The clinical value of the trained model depends on the delay between the availability of a measurement and the return of a prediction, which is a property of the deployment path. This path is therefore specified as part of the system. The trained network is exported to a portable computation graph and quantized to integer precision, which reduces memory footprint and arithmetic cost on commodity hardware without retraining [26]. Quantization is applied to the released model after federated training and therefore does not affect the privacy guarantee. Inference is executed at the edge tier rather than in the cloud, so the wide-area link is not on the critical path.

The delay is decomposed into four separately measured stages, because an end-to-end figure within budget may still be dominated by a removable stage. Preprocessing constructs and normalizes the input tensors from the sensor stream; the forward pass evaluates the exported graph; the conformal stage constructs the prediction set from the logits using the global threshold; and a final stage converts the output into the format required by the consuming system. Reporting the stages separately also isolates the cost of the conformal guarantee.

Latency is characterized by its distribution rather than its mean, and the budget must be met at the 95th percentile, since a clinical service level is defined by its tail. The measurement configuration is fixed in advance: the integer-quantized graph is executed on a general-purpose CPU provider, representing an edge node without an accelerator, for one thousand timed runs after one hundred discarded warm-up runs, and each of the four stages is timed within every run. A CPU provider is used because an edge deployment cannot be assumed to have a GPU.

I. Network and Energy Instrumentation

Three of the four design objectives concern quantities that cannot be observed from training: the wide-area traffic of the hierarchy, the reliability of update delivery, and the energy consumed per round by each tier. These are properties of the deployed topology and are therefore measured on simulated models of that topology rather than estimated from parameter counts.

The network is modeled with the three tiers of Section III-B. The device-to-edge segment uses a wireless local-area link and the edge-to-cloud segment a point-to-point backhaul, so the asymmetry between frequent local exchange over a low-cost link and infrequent global exchange over a high-cost link is represented explicitly. Each round is simulated in both directions, the downlink broadcast of the global model and the uplink transmission of client updates. Traffic volume, delivery reliability and end-to-end delay are recorded per round for the proposed configuration and for the flat and centralized baselines. Per-round delay quantifies the latency benefit of the hierarchy during training, which is distinct from the inference latency of the deployment path in Section III-H. Because the payload is the compressed update of Eqs. (10)–(11), the measured traffic is that of the specified system rather than of an uncompressed approximation.

Energy is modeled per tier from the power characteristics of the hardware and the completion time of the assigned work. The device tier represents constrained embedded nodes, the edge tier server-class accelerated hardware, and the cloud tier the aggregation host. Energy is reported per tier because a hierarchical design redistributes work rather than reducing it, and an aggregate figure would not show whether a saving at one tier is offset by an increase at another.

The targets for these quantities were fixed in advance and are listed in Table I. The measured values are reported in Section V.

TABLE I
DESIGN OBJECTIVES AND THEIR PRE-REGISTERED TARGETS. THE PROBLEM SPECIFICATION FIXES A NUMERIC THRESHOLD FOR COMMUNICATION REDUCTION BUT ASKS ONLY THAT THE LATENCY BENEFIT OF THE HIERARCHY BE QUANTIFIED, SO O1B IS STATED AS A DIRECTION RATHER THAN A PERCENTAGE. MEASURED VALUES ARE REPORTED IN SECTION V.

	Quantity	Target
O1	Uplink traffic reduction vs. centralized	$\geq 50\%$
O1b	Training-round latency vs. centralized	improvement
O2a	Privacy budget, composed total	$\varepsilon \leq 1$
O2b	Accuracy loss under the budget	≤ 2 pp
O3a	Inference latency, 95th percentile	< 100 ms
O3b	Successful model updates	$> 99\%$
O4	Energy saving per round vs. centralized	$\geq 20\%$

J. Empirical Auditing of the Guarantee

A differential privacy guarantee is an upper bound on the influence of an individual. It does not show what an adversary achieves in practice, nor that a system with a weaker guarantee is exploitable. Membership inference provides this complementary measurement [4]. PP-HFM is audited by individual-level membership inference. Members are the patients in the clients' training splits and non-members are patients held out entirely. Both groups are scored by the same released model using the same procedure, so any separation between them reflects membership.

Two attacks of different strength are used. The first compares each patient's mean per-record loss with a global threshold [17]. Aggregation to the patient level before thresholding is essential: a per-record attack measures record membership, which a record-level guarantee already protects, whereas averaging over a patient's records converts weak per-record signals into a per-individual signal. The second is a likelihood-ratio attack [13] adapted so that the unit of membership is the patient. Shadow models are trained on independently sampled halves of the patient population, with all records of a patient assigned together, and each shadow is trained with the same recipe as the target model. For each patient, the distribution of a logit-scaled confidence statistic is estimated separately over the shadows that include and exclude that patient, and membership is scored by the ratio of the two densities. A

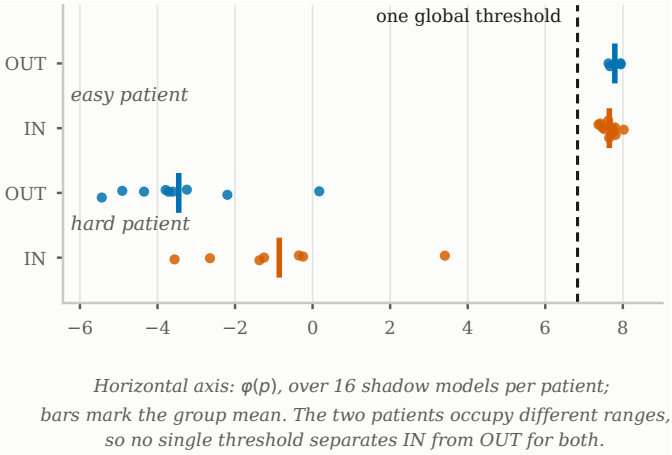


Fig. 5. Why per-patient calibration is necessary. Under a single global threshold an easy non-member can outscore a hard member, so membership signal is buried in per-patient difficulty. Fitting each patient's own in- and out-distributions from shadow models removes that confound and scores membership as a likelihood ratio.

global threshold is confounded by per-patient difficulty, so the attack strength it reports is a lower bound on leakage rather than an estimate [18]. Per-patient calibration removes this confound, as illustrated in Fig. 5.

Attacks are evaluated by the area under the receiver operating characteristic curve and by the true-positive rate at low false-positive rates, because an attack close to chance on average may still identify a few individuals with high confidence. When the number of non-members is too small to resolve a given false-positive rate, the value is reported as undefined rather than interpolated.

A negative audit result is informative only if the attack is capable of detecting leakage. The same procedure is therefore applied to a configuration known to memorize, namely non-private training of the same head over the same frozen encoders, and is required to detect clearly elevated leakage there. This positive control is a precondition of the audit and is reported with every result it qualifies.

K. Training Objective and Privacy Accounting

Let K clients be indexed by k, with client k holding a local dataset $\mathcal{D}_k$ of n_k records and $n = \sum_k n_k$. Clients are assigned to edges by a fixed map $e(\cdot)$. The records of client k are partitioned by individual into a set of patients $\mathcal{P}_k$, and W_p denotes the records contributed by patient p, with $k_p = |W_p|$; the partition is constructed so that W_p never spans two clients or two evaluation splits. Model parameters are written $\theta = (\theta_{\text{enc}}, \theta_{\text{head}})$, and ℓ is the class-weighted cross-entropy. The notation is summarized in Table II.

The learning objective is the standard weighted empirical risk over clients,

$$\min_{\theta} \; F(\theta) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \qquad F_k(\theta) = \frac{1}{n_k} \sum_{(x,y)\in\mathcal{D}_k} \ell\big(f_\theta(x), y\big), \tag{1}$$

TABLE II
NOTATION.

Symbol	Meaning
K, E	number of clients, of edge aggregators
$\mathcal{P}_k$, W_p	patients at client k; records of patient p
$k_p = \lvert W_p \rvert$	records contributed by patient p
R, τ_e	cloud rounds; edge rounds per cloud round
B	patients sampled per optimization step
C, σ	clipping norm; noise multiplier
q	Poisson sampling rate over patients
(ε, δ)	privacy budget
α	conformal miscoverage level
θ_{enc}, θ_{head}	encoder, fusion-head parameters

minimized only over θ_{head}, since θ_{enc} is pretrained on a disjoint public population and held fixed,

$$\theta_{\text{enc}} \text{ fixed}, \qquad \theta_{\text{head}} \leftarrow \text{privately optimized}. \tag{2}$$

At each optimization step client k draws a subset $S \subseteq \mathcal{P}_k$ by including each patient independently with probability

$$q = \frac{B}{|\mathcal{P}_k|}, \tag{3}$$

a rate defined over patients rather than records, which makes the accounted quantity patient-level. For each sampled patient the record gradients are accumulated and the sum is bounded,

$$g_p = \sum_{(x,y)\in W_p} \nabla_{\theta_{\text{head}}} \ell\big(f_\theta(x), y\big), \qquad \bar{g}_p = g_p \cdot \min\left(1, \frac{C}{\|g_p\|_2}\right). \tag{4}$$

Equation (4) determines the unit of the guarantee: because the bound is applied to the sum, the influence of patient p is at most C irrespective of k_p, whereas bounding each record separately would admit an influence of $k_p C$. A single perturbation scaled to the same norm is then applied to the aggregate,

$$\tilde{g} = \frac{1}{|S|}\left(\sum_{p\in S} \bar{g}_p + \mathcal{N}\big(0, \sigma^2 C^2 I\big)\right), \tag{5}$$

and the head is updated by $\theta_{\text{head}} \leftarrow \theta_{\text{head}} - \eta\, \tilde{g}$.

The noise multiplier is the smallest value satisfying the budget for every client simultaneously,

$$\sigma^\star = \min\Big\{\sigma : \max_k \; \varepsilon_k\big(\sigma, q_k, T_k\big) \le \varepsilon_{\text{train}}\Big\}, \tag{6}$$

where ε_k is obtained from a Rényi accountant [16] and the step count is

$$T_k = R\,\tau_e \left\lceil \frac{n_k}{B} \right\rceil. \tag{7}$$

The maximum in Eq. (6) is taken over all clients, and the ceiling in Eq. (7) rounds upward; replacing either with a representative client or a downward rounding yields a budget that is violated without any observable symptom.

Aggregation applies the weighted average at each tier. Writing $\theta_k^{(r,s)}$ for the model of client k after edge round s of cloud round r,

$$\theta_e^{(r,s)} = \sum_{k:\, e(k)=e} \frac{n_k}{\sum_{j:e(j)=e} n_j}\, \theta_k^{(r,s)}, \tag{8}$$

$$\theta^{(r)} = \sum_{e=1}^{E} \frac{N_e}{\sum_{e'} N_{e'}}\, \theta_e^{(r,\tau_e)}, \qquad N_e = \sum_{k:e(k)=e} n_k. \tag{9}$$

Updates are encoded before transmission. Sparsification retains the κd largest-magnitude coordinates of an update Δ of dimension d,

$$[\Delta^{\mathrm{sp}}]_i = \begin{cases} [\Delta]_i, & i \in \arg\operatorname{top}_{\kappa d} |\Delta| \\ 0, & \text{otherwise,} \end{cases} \tag{10}$$

and the retained values are quantized uniformly to b bits with step $s = (v_{\max} - v_{\min})/(2^b - 1)$,

$$Q(v) = v_{\min} + \operatorname{round}\Big(\frac{v - v_{\min}}{s}\Big)\, s. \tag{11}$$

The residual $\Delta - \Delta^{\mathrm{sp}}$ is accumulated locally and added to the next round's update. Both operations act on the already-perturbed quantity of Eq. (5) and are therefore post-processing.

Conformal calibration uses the least-ambiguous-classifier score on a held-out split,

$$r_i = 1 - \hat{p}_\theta(y_i \mid x_i)\,, \tag{12}$$

whose perturbed quantile at level $\lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil / n_{\mathrm{cal}}$ is released, aggregated across tiers to a global threshold $\hat{q}$, and used to form the prediction set $\mathcal{C}(x) = \{y : 1 - \hat{p}_\theta(y \mid x) \le \hat{q}\}$. The budget spent by a client is the composition of the training and calibration terms,

$$\varepsilon_{\mathrm{total}} = \varepsilon_{\mathrm{train}} + \varepsilon_{\mathrm{conf}}, \quad \varepsilon_{\mathrm{train}} = 0.8\,\varepsilon, \quad \varepsilon_{\mathrm{conf}} = 0.2\,\varepsilon, \tag{13}$$

and it is $\varepsilon_{\mathrm{total}}$, not $\varepsilon_{\mathrm{train}}$, that the system reports.

Finally, the audit statistic of Section III-J is the logit-scaled confidence of a patient, averaged over that patient's records,

$$\varphi(p) = \frac{1}{|W_p|} \sum_{(x,y)\in W_p} \Big(z_y(x) - \log \sum_{j \neq y} e^{z_j(x)} \Big), \tag{14}$$

which is the numerically stable form of $\log\big(\hat{p}_y/(1-\hat{p}_y)\big)$, and membership is scored by the ratio of the fitted in- and out-densities,

$$\Lambda(p) = \log \frac{\mathcal{N}\big(\varphi(p);\, \mu_{\mathrm{in}}(p), \sigma^2_{\mathrm{in}}(p)\big)}{\mathcal{N}\big(\varphi(p);\, \mu_{\mathrm{out}}(p), \sigma^2_{\mathrm{out}}(p)\big)}. \tag{15}$$

The objective in Eq. (1) is minimized over the fusion head only, using the mechanism of Eqs. (3)–(5) with the noise scale given by Eq. (6), while the encoders remain frozen. Optimization uses stochastic gradient descent on the perturbed gradient. The reported model is selected by validation performance, and the composed budget of Eq. (13) is reported instead of the training term alone.

IV. Experimental Setup

A. Datasets

The framework is evaluated on two publicly available multimodal corpora: VitalDB [21], a clinical monitoring collection of non-cardiac surgical cases, and WEAR [22], an outdoor activity collection combining body-worn inertial sensing with egocentric video. In both corpora, the two modalities originate from the same individual over the same interval and are paired by the recording apparatus rather than by a construction rule. This property is necessary for a claim about fusion, since a fusion benefit under rule-based pairing may reflect the rule rather than complementary information. The two corpora also cover the two deployment regimes relevant to the privacy mechanism: a client holding many individuals and a client holding one.

In VitalDB, modality A is the intraoperative monitoring record with eighteen channels, namely nine numeric physiological trends and nine binary indicators of whether each trend was recorded. Modality B is a thirty-four-dimensional vector of preoperative clinical and laboratory variables. All channels of a case originate from the same patient during the same operation and are time-aligned at capture across the waveform digitizer, the patient monitor and the anesthesia machine. The prediction target is postoperative admission to intensive care, which is positive in $19.65\,\%$ of windows. In WEAR, modality A consists of twelve inertial channels and modality B is a 2048-dimensional two-stream video feature computed over the same interval and synchronized in hardware. The task is activity recognition over eighteen classes.

Each corpus is partitioned into disjoint training, validation and test subsets, which are fixed across all experiments, so no record appears in more than one subset and all arms use the same split. For VitalDB, the partition contains $34{,}044$ training, $7{,}308$ validation and $7{,}360$ test windows from $6{,}257$ operations, distributed across nine clients by surgical specialty and assigned to two edge aggregators; an individual contributes 7.8 windows on average. For WEAR, the partition contains $26{,}424$ training, $5{,}395$ validation and $5{,}969$ test windows across twenty-two clients, one per participant, also assigned to two edges. Partitioning by specialty rather than at random produces realistic heterogeneity, since specialties differ in case mix, outcome prevalence and routinely recorded channels (Fig. 6). The validation subset is used only for model selection, and the test subset only for final evaluation.

Four controls are applied in preparing the clinical corpus, each preventing an inflation of the reported results that would not be visible during training. First, data are split by case, so all windows of an operation fall in the same subset; a window-level split would place highly correlated near-duplicates of training examples in the test set. Second, modality B is restricted to preoperative variables, and all intraoperative columns are excluded, because interventions such as transfusion or vasopressor administration are given in response to deterioration and therefore encode the outcome. Third, missing values are not simply imputed: gaps are filled within a case and paired with an explicit binary indicator, because, for example, an arterial line is placed only in patients

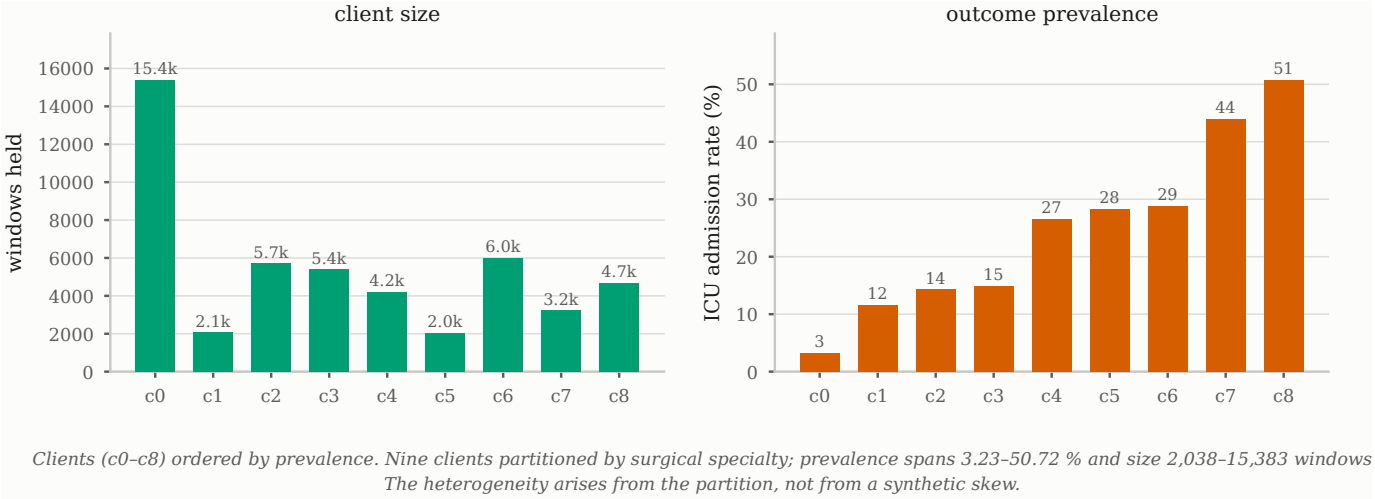


Fig. 6. Heterogeneity of the clinical partition. Partitioning by surgical specialty rather than at random yields clients that differ by a factor of seven in size and by more than fifteen in outcome prevalence, from $3.23\,\%$ to $50.72\,\%$. The skew is a property of how care is organized, not an artifact imposed for the experiment.

judged to be at risk, so its absence carries information about acuity. Fourth, normalization statistics are computed over training cases only, with estimators that handle missing values, because a naive estimator returns an undefined statistic for a fully missing channel and propagates it to every window. In addition, a separate set of individuals, disjoint at the individual level from all client data, is reserved before federated training and used only to pretrain the encoders, so that the reported budget holds end to end.

B. Implementation Details

All arms use the same training configuration, so that differences can be attributed to the mechanism under test. Each client performs one local pass per federated call, five edge rounds are executed per cloud round, and training runs for ten cloud rounds. Optimization is stochastic gradient descent on the privatized gradient, with a per-individual clipping norm of one and the noise multiplier obtained from the calibration of (6) over all clients. Updates are sparsified to the largest twenty percent of coordinates and quantized to eight bits before transmission, with error feedback of the residual. Conformal calibration is performed once per run at a miscoverage level of 0.1, and its cost is included in the reported budget according to (13). The encoders are pretrained centrally and then frozen, so the privatized parameters are those of the fusion head only: $559{,}874$ of $1{,}864{,}227$ parameters, or $30.0\,\%$.

Unless stated otherwise, all results, including the system-level objectives, the privacy–utility comparisons and the membership-inference audit, were obtained with the *oracle* encoder configuration of Section III-C, in which the encoders were pretrained on the pooled client training data. This configuration does not provide a valid end-to-end guarantee. It is used because it isolates the behavior of the privatized head from the origin of the representation, and because using the same encoders in every arm keeps the comparisons between arms controlled. The *disjoint* configuration, which provides the guarantee, is evaluated separately, and the difference is reported in Section V-A as the cost of a valid guarantee. Absolute utilities should therefore be read as upper bounds for the deployable configuration, whereas differences between arms, on which the claims rely, are unaffected. All normalization layers are group or layer normalizations, as explained in Section III-C.

Models are implemented in PyTorch 2.5, with per-sample gradients and Rényi accounting provided by Opacus 1.5. Deployment models are exported to ONNX and quantized to eight-bit integer precision, and inference is measured with ONNX Runtime on a general-purpose CPU provider, since an edge node cannot be assumed to have an accelerator. Network behavior is simulated in ns-3 3.46, with the device-to-edge segment over a wireless local-area link and the edge-to-cloud segment over a point-to-point backhaul. Energy is modeled in CloudSim Plus separately for the device, edge and cloud tiers. Training was performed on a single workstation with one NVIDIA RTX A2000 GPU.

C. Evaluation Metrics

Classification performance is assessed with two complementary measures. The area under the receiver operating characteristic curve summarizes ranking performance across all decision thresholds. Average precision summarizes the precision–recall trade-off,

$$\mathrm{AP} = \sum_{n} \left(R_n - R_{n-1}\right) P_n, \tag{16}$$

where P_n and R_n are the precision and recall at the nth threshold. Average precision is reported because the clinical task is imbalanced, with $19.65\,\%$ positive windows. Under this imbalance, accuracy is nearly uninformative, since a constant negative prediction achieves about four-fifths accuracy, and the area under the ROC curve is optimistic. Average precision reflects performance on the minority class.

Metrics are also aggregated per individual. The model scores windows, but the clinical question concerns an oper-

ation, and windows of the same case are highly correlated, so window-level metrics overstate the effective sample size. The predictions of a case are therefore averaged,

$$\bar{p}_c = \frac{1}{|W_c|} \sum_{w \in W_c} p_w, \tag{17}$$

giving one prediction per individual. Case-level metrics are computed from $\bar{p}_c$ using the same expressions as for p_w.

Comparisons between arms are paired by seed, since the arms share their partition, initialization schedule and calibration data. For S paired seeds with per-seed difference d_s, the reported interval is

$$\bar{d} \pm t_{S-1,\,0.975} \frac{s_d}{\sqrt{S}}, \tag{18}$$

where s_d is the sample standard deviation of the differences. Student's t distribution is used instead of a normal approximation because the number of seeds is small. Seed counts are fixed before each comparison and are not increased after inspecting the results.

Communication, latency, reliability and energy are reported as defined in Section III-I, and every network quantity is measured on rounds in which the payload traversed the simulated topology.

D. Experimental Protocol

The proposed system is compared with differentially private federated averaging in three configurations, each isolating one design decision. The first trains the full parameter vector under the privacy budget, which is the conventional application of DP-SGD to a federated model. The second restricts private optimization to the fusion head over randomly initialized frozen encoders, isolating the effect of reducing the privatized dimension. The third restricts private optimization to the fusion head over *pretrained* frozen encoders. This arm controls for a confound: without it, the benefit of parameter-efficient private training cannot be separated from the benefit of the pretrained representation. A non-private model trained with an otherwise identical recipe provides the reference for the accuracy cost of the budget. All arms share the partition, schedule, compression settings and evaluation procedure, and differ only in the factor under test.

The same evaluation protocol is applied to every arm. The checkpoint with the highest validation performance is retained, without use of test data. Each retained checkpoint is evaluated once on the test subset, and all metrics are computed from the same predictions with the same aggregation.

The privacy guarantee is audited by individual-level membership inference using the two attacks of Section III-J. Each attack is also applied to a configuration known to memorize, namely non-private training of the same head over the same frozen encoders, as a positive control. A negative result is accepted only if the same procedure detects clearly elevated leakage on the control.

A further experiment separates the two sources of the cost of the individual-level unit. Moving the guarantee from the record to the individual both reduces the number of units sampled per step and bounds the summed contribution of each individual, and the aggregate comparison cannot separate the two. The number of units sampled per step is therefore held equal in both arms, which removes the first effect and leaves only the clipping geometry, and the number of training records retained per patient, k, is varied over $\{1, 2, 4, 8\}$ with three seeds at $\varepsilon = 1$. Subsampling is applied to the training partition only; validation and test sets are unchanged, so every configuration is evaluated on the same held-out data. Because reducing k also reduces the training set size, absolute utility decreases with k and is not comparable across k. The quantity of interest is the paired difference between arms at each k, where both arms use the same data, partition and seed. At $k = 1$, a patient contributes a single record, so the two arms must agree, which provides a falsification test of the design.

TABLE III
MEASURED OUTCOMES AGAINST THE PRE-REGISTERED TARGETS OF TABLE I. NETWORK QUANTITIES ARE MEASURED AT PACKET LEVEL ON EVERY SIMULATED ROUND OF BOTH DEPLOYMENTS. THE ACCURACY ROW IS A PAIRED COMPARISON AGAINST A NON-PRIVATE REFERENCE TRAINED UNDER THE IDENTICAL RECIPE; † MARKS AN INTERVAL THAT SPANS ITS THRESHOLD AND IS THEREFORE INCONCLUSIVE AT THE PRE-REGISTERED SAMPLE SIZE.

	Quantity	Clinical	Wearable	Target
O1	Uplink reduction	90.8%	96.1%	$\geq 50\%$
O1b	Round latency	16.9%	21.7%	improvement
O2a	Budget (composed)	0.970	0.966	$\varepsilon \leq 1$
O2b	Accuracy change	-0.92 pp	$1.97^{\dagger}$ pp	≤ 2 pp
O3a	Latency (P95)	4.58 ms	1.22 ms	< 100 ms
O3b	Update success	100.00%	99.94%	$> 99\%$
O4	Energy saving	72.3%	76.8%	$\geq 20\%$

V. RESULTS

A. System-Level Objectives

The framework is evaluated on two deployments that differ in client composition. In the clinical deployment, nine institutional clients hold the records of 6,257 patients, so each client contains many individuals, and each individual contributes 7.8 records on average. In the wearable deployment, each of the twenty-two clients corresponds to a single participant who contributes approximately 1,750 records. The second deployment also tests whether the system-level behavior generalizes across topology and payload size.

Table III reports the measured value of each design objective of Table I. All targets are met on the clinical deployment. On the wearable deployment, all targets are met except the accuracy bound, which is inconclusive.

All network quantities are measured on every simulated round. An earlier simulator configuration evaluated one round in ten at packet level and composed the remaining rounds analytically. The two approaches disagree: the analytical model placed the communication term of the centralized baseline at 256 ms, whereas the simulated network delivers 23.6 ms, which overstated the latency benefit of the hierarchy by a factor of about four. No analytically composed rounds contribute to the values reported here.

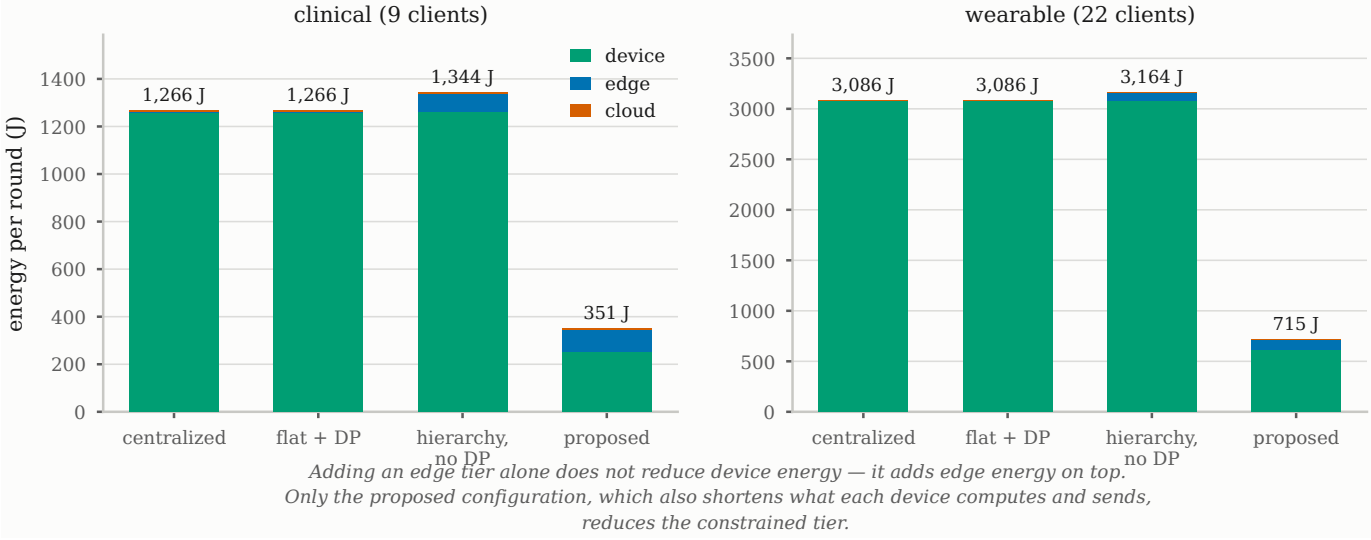


Fig. 7. Energy per training round, attributed by tier, on both deployments. The hierarchy alone redistributes work without reducing it: device energy is unchanged and edge energy is added. Only the proposed configuration reduces the energy of the constrained tier.

TABLE IV
INFERENCE LATENCY DECOMPOSITION, INTEGER-QUANTIZED GRAPH ON A CPU PROVIDER, 1000 TIMED EXECUTIONS AFTER 100 DISCARDED WARM-UP EXECUTIONS.

Stage	Mean (ms)	P95 (ms)
Preprocessing	0.140	0.147
Forward pass	4.221	4.395
Conformal set construction	0.030	0.037
Post-processing	0.011	0.014
Total	4.401	4.582

The reported privacy budget is the composed total. Training accounts for $\varepsilon = 0.7705$ and the release of the conformal calibration quantile for a further 0.2000, giving 0.970 against a budget of one. Reporting only the training term would understate the expenditure by $23\,\%$.

1) Communication and latency: Two-tier aggregation reduces uplink traffic by $90.8\,\%$ relative to centralized training, against a target of $50\,\%$. The measurement is taken from the deployed compressor and includes the index array that top-κ sparsification transmits with the retained values, which analytical payload estimates omit and which is not negligible at high sparsity.

The hierarchy also reduces the end-to-end latency of a training round from 90.5 ms to 75.2 ms in the clinical deployment, a reduction of $16.9\,\%$. The reduction is positive in every simulated round, ranging from $8.2\,\%$ to $25.8\,\%$. It is smaller than the traffic reduction because the communication term falls from 23.6 ms to 16.5 ms, while the compute terms (sensing, preprocessing, inference and aggregation) are common to all topologies and dominate the total. The hierarchy reduces link time, not computation.

In the wearable deployment, round latency falls from 104.5 ms to 81.8 ms, or $21.7\,\%$, with the communication term falling from 37.6 ms to 23.2 ms. The larger reduction is consistent with the larger topology: twenty-two stations contend for the same wireless medium rather than nine, so uplink latency is a larger share of the total. The latency benefit of the hierarchy therefore depends on the topology.

2) Cost of a valid guarantee: The results above were obtained with the oracle encoder configuration, in which the frozen representation was pretrained on the pooled client training data. This configuration does not provide a valid end-to-end guarantee, and its utility is therefore an upper bound. With the disjoint configuration, in which the encoders are pretrained only on a population reserved before federated training and disjoint from every client, the area under the ROC curve decreases by 0.0322 $(0.8429 \to 0.8107)$, by 0.0310 at the case level, and average precision by 0.0763. This difference is the cost of an end-to-end valid guarantee. Comparisons between arms are unaffected, because all arms share the same encoders.

3) Privacy and utility: The composed budget of 0.970 satisfies the requirement $\varepsilon \leq 1$. Relative to a non-private reference trained with the same recipe, the change in balanced accuracy is -0.92 pp, with a paired $95\,\%$ confidence interval of $[-2.82, +0.99]$ over five seeds. The interval contains zero, so no accuracy cost is detectable at this sample size. The requirement of at most 2 pp is met.

On the wearable deployment, the same comparison gives 1.967 pp with an interval of $[1.760, 2.173]$ over fifteen seeds. The interval spans the 2 pp threshold, so this objective is *inconclusive*. The seed count was fixed in advance and was not increased after inspecting the interval.

4) Quality of service: Table IV decomposes the inference latency. Construction of the conformal prediction set, which provides the coverage guarantee, takes 0.030 ms of the 4.401 ms mean inference time, or $0.68\,\%$, and 0.037 ms at the 95th percentile. The differentially private mechanism adds no inference cost, because its noise is applied to gradients during training and changes only the parameter values. The conformal set is therefore the only guarantee-related component with an inference cost, at less than one percent of the total.

The 95th-percentile latency is 4.58 ms in the clinical deployment and 1.22 ms in the wearable deployment, against a budget of 100 ms, both measured on the integer-quantized graph on a CPU provider. This twentyfold margin is due to the compact fusion head and integer arithmetic, not to the hierarchy or the privacy mechanism. The difference between the deployments follows from the input dimension. The margin allows a larger backbone to be used within the same latency budget.

Update delivery succeeds for $100.00\,\%$ of transmissions in the clinical deployment and $99.94\,\%$ in the wearable deployment, against a target of $99\,\%$. Both values are measured on every round. Under the earlier configuration, in which nine rounds in ten were composed analytically, the clinical value was $99.27\,\%$. The wearable deployment does not reach perfect delivery, with a per-round minimum of $99.15\,\%$, as expected when twenty-two stations share one medium.

5) Energy: Energy per round decreases from 1266.2 J to 350.8 J in the clinical deployment, a saving of $72.3\,\%$, and from 3086.2 J to 714.8 J in the wearable deployment, a saving of $76.8\,\%$, against a target of $20\,\%$. Energy is reported per tier because a hierarchy redistributes computation rather than eliminating it (Fig. 7). Introducing the edge tier *alone* leaves device energy unchanged and adds edge energy, so the total increases from 1266 J to 1344 J. The saving occurs only in the configuration that also reduces device-side computation and transmission, which lowers the device term fivefold.

B. The Privacy Unit and the Budget

Two comparisons quantify the cost of the design decisions of Section III-D. Both are paired by seed against an arm that differs only in the factor under test.

1) Cost of moving the unit from the record to the individual: Enforcing the guarantee over individuals rather than records reduces the area under the ROC curve by 0.0188, with a paired $95\,\%$ interval of $[-0.0255, -0.0120]$ over five seeds. The case-level change is -0.0193 $[-0.0266, -0.0120]$, and average precision decreases by 0.0783 $[-0.1046, -0.0520]$. These intervals exclude zero. The cost is larger in average precision than in the area under the ROC curve, consistent with the minority class being harder to fit under a coarser sampling unit.

Two effects contribute to this cost. Sampling patients rather than records reduces the number of effective units per step from $34{,}044$ to $4{,}372$ on this partition, and clipping a patient's summed gradient constrains individuals with many records more strongly than those with few. Section V-B3 isolates the second effect by holding the number of units per step equal in both arms.

2) Initialization, not the privatized subset: The proposed recipe differs from conventional private training in two respects: it starts from pretrained encoders, and it privatizes only the fusion head. A two-arm comparison cannot separate these factors, so three arms are compared. B4 is full-model private training from a random initialization, the prior-art configuration. B7 is full-model private training from the pretrained initialization, which changes only the initialization. B6 is head-only private training from the pretrained initialization, which changes only the privatized subset. Fig. 8 shows the three arms on the clinical corpus, and Table V reports both steps on both deployments over three seeds.

TABLE V
DECOMPOSITION OF THE RECIPE IN MACRO AUROC, THREE SEEDS. THE INITIALIZATION STEP B4→B7 HOLDS THE PRIVATIZED SUBSET AT THE WHOLE NETWORK AND CHANGES ONLY THE STARTING POINT; THE SUBSET STEP B7→B6 HOLDS THE STARTING POINT AND PRIVATIZES $30\,\%$ OF THE NETWORK INSTEAD OF ALL OF IT, AND IS REPORTED AS MEAN $\pm$ A STUDENT-t HALF-WIDTH BECAUSE IT IS THE STEP THE CLAIM TURNS ON.

	Clinical		Wearable	
ε	B4→B7	B7→B6	B4→B7	B7→B6
∞	$+0.0187$	$+0.0036 \pm 0.0329$	$+0.0021$	-0.0002 ± 0.0003
8	$+0.0649$	$+0.0014 \pm 0.0084$	$+0.0883$	$+0.0001 \pm 0.0002$
4	$+0.0803$	$+0.0033 \pm 0.0049$	$+0.1257$	$+0.0001 \pm 0.0002$
2	$+0.0985$	$+0.0031 \pm 0.0063$	$+0.1918$	$+0.0002 \pm 0.0001$
1	$+0.1221$	$+0.0026 \pm 0.0122$	$+0.2827$	$+0.0003 \pm 0.0008$
0.5	$+0.1447$	$+0.0030 \pm 0.0204$	$+0.3580$	$+0.0013 \pm 0.0088$

Almost the entire effect is due to the initialization, and it increases as the budget tightens. On the clinical corpus, the change from random to pretrained initialization is worth 0.0187 without noise and 0.1447 at the tightest budget, increasing monotonically in between; on the wearable corpus it ranges from 0.0021 to 0.3580. Restricting the privatized subset is worth at most 0.0036 on the clinical corpus and 0.0013 on the wearable corpus at any budget, and its intervals cover zero throughout.

This result contradicts a dimensionality explanation. If the advantage arose because a smaller parameter vector absorbs less noise, B7, which privatizes all $1{,}864{,}227$ parameters, should perform worse than B6, which privatizes $559{,}874$, and the difference should be largest at the tightest budget. At $\varepsilon = 0.5$, however, the two arms differ by only 0.0030, with an interval covering zero, although B7 privatizes 3.3 times as many parameters under the same budget. This is consistent with the noise scale being determined by the budget, the number of steps and the sampling rate rather than by the number of parameters.

The same pattern appears in sensitivity to the budget. Across the full range, the randomly initialized arm loses 0.1181 on the clinical corpus and 0.3716 on the wearable corpus, where it falls to 0.6247, close to chance. Both pretrained arms change by less than 0.016 on either deployment. Robustness to a tight budget is therefore determined by the initialization. Under a fixed budget, a legitimate pretrained initialization should take priority over the choice of privatized parameters.

Restricting the privatized subset does reduce communication. Only head parameters are transmitted, which produces the uplink reductions reported in Section V-A, and Table V shows that this comes at no measurable cost in accuracy. It is not, however, an accuracy mechanism, and a two-arm comparison of head-only training from a pretrained initialization against full-model training from a random initialization does not support such a claim, because the two arms differ in both respects.

3) Records per individual: Because a patient-level guarantee bounds the *summed* contribution of an individual, its

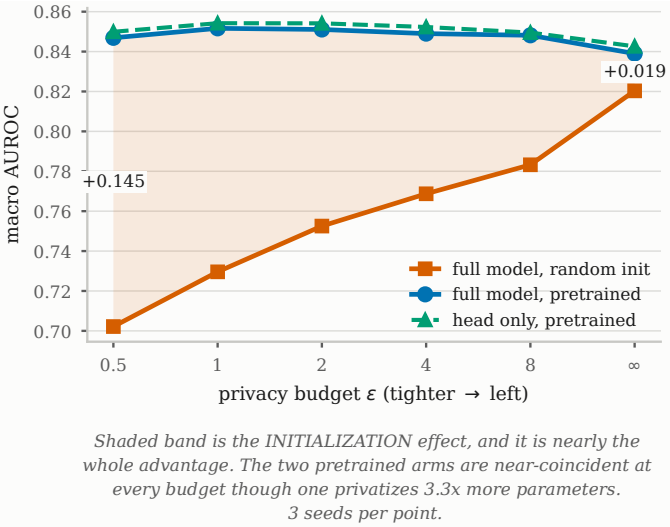


Fig. 8. Decomposing the recipe across the budget on the clinical corpus. The shaded band is the initialization effect, obtained with the privatized subset held at the whole network, and it is nearly the entire advantage. The two pretrained arms nearly coincide at every budget, although one privatizes 3.3 times as many parameters as the other, which excludes an explanation based on privatized dimensionality. Three seeds per point.

TABLE VI
AVERAGE PRECISION BY RECORDS PER INDIVIDUAL k. THE GAP IS PAIRED BY SEED AND REPORTED AS MEAN $\pm$ THE HALF-WIDTH OF A STUDENT-t INTERVAL OVER THREE SEEDS; AN INTERVAL EXCLUDING ZERO IS ONE WHOSE HALF-WIDTH IS SMALLER THAN ITS MEAN.

k	Record	Individual	Gap	p
1	0.5015	0.5067	-0.0052 ± 0.0613	0.7514
2	0.5590	0.5084	0.0507 ± 0.0090	0.0017
4	0.5924	0.5092	0.0832 ± 0.0138	0.0015
8	0.5955	0.5089	0.0866 ± 0.0263	0.0049

cost is expected to increase with the number of records per individual and to vanish when each individual contributes a single record. Table VI reports both arms as k, the number of training records retained per patient, is varied over $\{1, 2, 4, 8\}$ with three seeds. The number of units sampled per step is equal in both arms, which removes the sampling-rate effect, so only the clipping geometry differs. The validation and test partitions are unchanged.

At $k = 1$, the two arms are indistinguishable, with a gap of -0.0052 and an interval covering zero, as required by the construction. The divergence at larger k can therefore be attributed to the privacy unit rather than to the experimental setup. From $k = 2$ onward, the gap is resolved and increases monotonically, reaching 0.0866 at $k = 8$; the increment from $k = 4$ to $k = 8$ is an order of magnitude smaller than the preceding one.

The individual-level arm remains flat in k, within $[0.4967, 0.5176]$ across all configurations, whereas the record-level arm increases from 0.5015 to 0.5955. The widening gap therefore reflects a lack of improvement under the stricter unit rather than a degradation. Additional records per patient provide signal that the record-level arm exploits and the individual-level arm cannot, because the bound on the summed contribution is tightest for the individuals who contribute the additional records. On this corpus, collecting more data per patient therefore increases the relative cost of an individual-level guarantee.

TABLE VII
AVERAGE PRECISION BY RECORDS PER INDIVIDUAL k ON THE EXTERNAL CORPUS, THREE SEEDS, PAIRED BY SEED. INTERVALS ARE STUDENT-t HALF-WIDTHS AS ELSEWHERE. THE ROW $k = 16$ LIES BEYOND THE RANGE OF THE FIRST CORPUS. THE FINAL COLUMN IS THE VALUE PREDICTED BY EQ. (1), FIXED BEFORE THE CORPUS WAS BUILT.

k	Record	Individual	Gap	p	Predicted
1	0.2105	0.2212	-0.0107 ± 0.0211	0.1594	0.0000
2	0.2245	0.2185	$+0.0060 \pm 0.0100$	0.1215	0.0515
4	0.2318	0.2208	$+0.0110 \pm 0.0714$	0.5746	0.0816
8	0.2374	0.2239	$+0.0135 \pm 0.0592$	0.4296	0.0876
16	0.2363	0.2258	$+0.0105 \pm 0.0354$	0.3310	0.0878

The area under the ROC curve changes in the same direction, by 0.0006, 0.0094, 0.0133 and 0.0132 over the same values of k, but every interval covers zero at three seeds. The effect is therefore resolved in average precision only. Accuracy is not reported for these arms, because an uncalibrated decision threshold on an imbalanced corpus keeps it near the majority rate; an area under the ROC curve of approximately 0.835 confirms that the models rank well.

4) External validation of the records-per-individual law: The law of Section V-B3 was measured on a single corpus whose clients are surgical specialties rather than institutions, so it is not established whether it characterizes patient-level private federated learning in general or only that cohort. It is therefore tested on a second clinical corpus of different construction: 3,782,805 hourly windows from 132,736 individuals across 187 clients that are *real hospitals*, with a median of 444 individuals per client and 28.50 records per individual. The label is in-unit mortality, with a prevalence of 0.0422.

Two properties of this replication are relevant. First, its encoders were pretrained on 526,867 windows verified to be disjoint from every client's training data, so the results do not carry the oracle-configuration caveat that applies elsewhere in this paper. Second, the prediction was fixed before the corpus was constructed. Fitting the four VitalDB points of Table VI gives

$$\mathrm{gap}(k) = A\left(1 - e^{-b(k-1)}\right), \tag{1}$$

with $A = 0.0878 \pm 0.0035$ and $b = 0.883 \pm 0.129$ at a root-mean-square error of 0.0028. This curve, together with an acceptance band of ± 0.030 derived from the VitalDB seed spread, was registered before any record of the second corpus was accessed.

The control holds. At $k = 1$, the gap is -0.0107, with an interval covering zero and a magnitude within the pre-specified tolerance of 0.020. The two arms are therefore not separated where the construction requires them to agree, and the remaining rows can be attributed to the privacy unit.

The point forecasts are not confirmed. At every $k > 1$, the gap lies outside the ± 0.030 band, and the measured cost is approximately one sixth of the predicted cost: 0.0135 against 0.0876 at $k = 8$.

Two properties of the law are reproduced. The first is the ordering: the gap is negative at $k = 1$, positive at $k = 2$, and increases through $k = 4$ to $k = 8$, as described by Eq. (1). This is consistent with the mechanism, since the bound on an individual's summed contribution is tightest for individuals who contribute additional records.

The second is the saturation, which the first corpus cannot test. The registered prediction was that the gap beyond $k = 8$ would exceed the gap at $k = 8$ by less than 0.015. Across paired seeds, the increment from $k = 8$ to $k = 16$ is -0.0030, with an interval of $[-0.0316, +0.0255]$ ($p = 0.692$). The gap does not continue to increase, and the registered bound is met. The cost of the guarantee therefore approaches a ceiling in k on both corpora.

The asymmetry between the arms is also reproduced. Across all k, the individual-level arm varies within $[0.2185, 0.2258]$, a range of 0.0073 against the pre-specified 0.020, whereas the record-level arm increases from 0.2105 to 0.2374. As on the first corpus, the widening gap reflects improvement of the record-level arm rather than degradation of the individual-level arm. The corpora therefore differ in the magnitude of the effect but not in its mechanism, and, as registered in advance, the shape is interpreted as structural and the magnitude as corpus-dependent.

The statistical power of this replication is limited. At three seeds, no gap in Table VII is resolved: every interval covers zero, and no p-value is below 0.12. The seed standard deviation of the gap at $k = 8$ is of the same order as the gap, and resolving an effect of this size at conventional power would require approximately fifty seeds, against the three fixed in advance. The seed count was not increased after inspecting the results. The trend is therefore reported as directionally consistent with the VitalDB law and unresolved on this corpus. The area under the ROC curve is less informative: its gaps over the same values of k are -0.0012, -0.0015, $+0.0018$, $+0.0001$ and $+0.0105$, with no trend over the range covered by the first corpus. Average precision is the primary measure on this corpus, as on the first, and the area under the ROC curve provides no evidence either way.

One possible explanation for a preserved shape at a lower ceiling is offered as a hypothesis. The two tasks differ in attainable precision by approximately a factor of two: the arms on this corpus lie within $[0.1960, 0.2446]$ at a prevalence of 0.0422, whereas those on the first corpus lie between 0.4967 and 0.5955. A gap in average precision is bounded by the attainable precision, so a cost that is a fixed fraction of attainable performance would produce a smaller absolute gap on the harder task with the same ordering in k. Under this explanation, the transferable quantity is the cost relative to attainable precision rather than the ceiling A. Distinguishing this from a corpus-specific ceiling requires a third corpus.

The range of k is limited by the corpus. The registered design specified k up to 32. On this corpus, however, a budget of twenty-four windows per admission means that only $20.2\,\%$ of individuals reach thirty-two records, and these individuals have multiple admissions and are not representative of the cohort. Estimating the arm at $k = 32$ would therefore measure the effect of readmission rather than of the privacy unit.

TABLE VIII
AVERAGE PRECISION GAP (RECORD ARM MINUS INDIVIDUAL ARM, PAIRED BY SEED, THREE SEEDS) ACROSS THE THREE GRIDS. THE FIRST TWO DIFFER ONLY IN THE TRAINING BUDGET. THE FORECAST COLUMN IS EQ. (1), FITTED ON THE FIRST COLUMN AND NEVER REFITTED.

k	Forecast	First corpus $\varepsilon_{\mathrm{tr}} = 1.00$	First corpus $\varepsilon_{\mathrm{tr}} = 0.80$	Second corpus $\varepsilon_{\mathrm{tr}} = 0.80$
1	0.0000	-0.0052	$+0.0380$	-0.0107
2	0.0515	$+0.0507$	$+0.0355$	$+0.0060$
4	0.0816	$+0.0832$	$+0.0945$	$+0.0110$
8	0.0876	$+0.0866$	$+0.0967$	$+0.0135$

The maximum was reduced to 16, which retains $91.0\,\%$ of individuals, and $k = 32$ is reported as untested. This change was recorded as a dated amendment to the registration before any result beyond $k = 8$ was computed, and it did not alter any predicted value.

5) A matched-budget control for the external test: The comparison of the two preceding subsubsections is confounded by the training budget. The curve of Eq. (1) was *fitted* on a first-corpus grid trained at $\varepsilon_{\mathrm{train}} = 1.00$ and *tested* on a second-corpus grid trained at $\varepsilon_{\mathrm{train}} = 0.80$, the operating point of Section IV. Since additional noise reduces the difference between any two arms, which is the same direction as the observed shortfall, the shortfall cannot be attributed to the corpus without a control.

The pre-registered correction re-runs the first corpus at the operating point of the second, with k, seeds, partitions and protocol unchanged, and without refitting the forecast. Table VIII compares the three grids.

Taken at face value, the budget explains almost none of the shortfall. The matched gaps are not smaller than the original ones: at $k = 8$, the matched gap is $+0.0967$ against $+0.0866$, so the lower budget moved the first corpus in the *opposite* direction to that predicted by the confound. Decomposing the total change at $k = 8$ attributes $+0.0101$ to the budget and -0.0832 to the corpus.

Two qualifications apply. First, the re-run did not change the budget alone: the operating point also changes the batch size from 32 to 64 units, so the budget term is the joint contribution of the budget and the sampling configuration. Second, the matched grid fails its own positive control. At $k = 1$, where the two arms must agree by construction, it separates them by $+0.0380$, with all three seeds of the same sign, against a pre-specified tolerance of 0.020. The interval covers zero, so the result is formally unresolved, but the consistent sign is notable, and both the original grid (-0.0052) and the external grid (-0.0107) pass the same check.

The failure is attributable to the sampling configuration rather than to the budget. At $k = 1$, the first corpus retains one record per individual, giving a training pool of $4{,}372$ records, the smallest in this study; with a batch of 64, this yields approximately sixty-eight optimization steps per round for either arm. The original grid uses the same pool with a batch of 32 and its control passes, and the external corpus uses $76{,}589$ records with a batch of 64, seventeen times more data at the same batch size, and its control also passes. The failing configuration combines the smallest pool with the

largest batch, which indicates a lack of power rather than an effect of the privacy unit, consistent with the width of the interval.

Subtracting each grid's own $k = 1$ offset from its other rows gives a conservative estimate. At $k = 8$, the budget and sampling term then accounts for -0.0331 and the corpus for -0.0345; at $k = 4$, for -0.0320 and -0.0347. The two contributions are then comparable. Under the face-value estimate, the budget contributes nothing; under the conservative estimate, it contributes about half. Under neither does it account for the factor of six separating the two corpora at $k = 8$, and under neither does the corpus term change sign. The matched grid therefore shows that the budget mismatch cannot account for the external shortfall, but not that the mismatch contributed nothing.

The control failure is confined to this re-run. The scaling law of Table VI and its external test in Table VII are measured on grids whose $k = 1$ controls pass, so neither depends on the grid reported here; only the decomposition of the difference between them does. The matched grid is therefore reported as a bound on a confound rather than as a measurement.

6) Is the cost an artifact of the clipping rule?: The individual-level arm clips each individual's *summed* gradient to a fixed norm, so clipping is strongest for individuals with many records. The flatness of this arm in k could therefore result from the clipping rule rather than from the guarantee. In that case, it should be partly recoverable by dividing the summed gradient by the record count, or by its square root, before clipping. Neither normalization changes the guarantee: sensitivity is determined by the clipping norm, so the noise scale and the budget are identical under all three rules, and any difference is due to optimization.

The premise holds in direction but not in magnitude. On the external corpus, the fraction of individuals whose gradient is clipped increases from 0.148 for those with one to four records to 0.502 for those with forty-nine or more, but the rank correlation between record count and clipping is only 0.138. Normalizing by the count halves the clipped fraction, from 0.3945 to 0.2039, rather than eliminating it. The first corpus cannot test this question, because nearly every individual contributes exactly eight records, so normalization by the count is a uniform rescaling. The test is therefore performed on the external corpus only, using its natural partition rather than the capped partitions of the scaling grid, since capping would remove the variation in records per individual, which ranges from two to 192.

Normalization does not recover utility. At the operating budget and three pre-specified seeds, the individual-level arm attains an average precision of 0.2266 with the fixed bound, 0.2265 with normalization by the count, and 0.2242 with normalization by its square root. The paired differences relative to the fixed bound are -0.0001 ± 0.0087 and -0.0024 ± 0.0095. The pre-registered threshold for recovery was $+0.010$ for normalization by the count, obtained by scaling the threshold for the first corpus to the precision attainable on this corpus. The one-sided $95\,\%$ upper bound on the difference is $+0.0058$, so an effect of the registered size is excluded. The area under the ROC curve changes in the same direction, by -0.0022 and -0.0011. The square-root rule, predicted to fall between the other two, falls below both, although by less than the seed variation.

The intervention was effective: from the third round onward, on every seed, normalization by the count produced the lowest clipped fraction and the fixed bound the highest. In the final round, the fractions are 0.157 against 0.212 on one seed, and 0.046 against 0.086 and 0.047 against 0.118 on the others. On one seed, the selected checkpoint for every rule is the first round, before the rules diverge, so the three rules agree to within 10^{-4} and this seed carries little information about the contrast. The cost of the individual-level unit is therefore not an artifact of the clipping rule.

C. Does Fusion Earn Its Place?

The benefit of combining the two modalities is evaluated separately from the federated system, because the question concerns the data rather than the protocol, and federation and privacy noise would add variance and confounds. Each arm is retrained on its own inputs rather than evaluated with one modality zeroed, since zeroing a stream of a jointly trained model measures robustness to a missing stream rather than the information it carries.

Over five seeds, the fused model attains 0.8653 ± 0.0072 in area under the ROC curve, against 0.8283 ± 0.0087 for the better single modality and 0.7680 ± 0.0045 for the weaker, where each interval is a Student-t half-width. The gain of 0.0370 is 5.9 times the seed standard deviation, and the gain in average precision is 0.0671. The modalities are paired by the recording apparatus, from the same patient during the same operation, so the gain does not arise from a pairing rule. With two seeds, the same experiment gave gains of 0.0411 and 0.1062, at 14.1 times the seed standard deviation; with five seeds, the average-precision gain is $37\,\%$ lower and the effect size less than half as large. Only the five-seed values are used.

On the wearable deployment, the better single modality already attains 0.9976, leaving a fusion gain of 0.0004 and no room for a second stream to contribute. The benefit of fusion therefore depends on the task and must be established per task.

1) The external clinical corpus, and the condition that separates the three: On the external clinical corpus, which has more headroom than the wearable corpus, fusion yields no gain. The fused model attains 0.8249 in area under the ROC curve against 0.8212 for the admission-record stream alone, a gain of 0.0037, and -0.0001 in average precision. The vital-sign stream alone reaches 0.6748, so the two streams differ substantially in strength. This motivates a second hypothesis, that fusion requires the streams to be of comparable strength, which was pre-registered before testing.

The test weakens the stronger stream until the two are balanced and measures whether a gain appears. Zeroing the dominant admission feature reversed the dominance rather than balancing the streams (Table IX). Scaling the feature was ineffective, because the stream is standardized and its encoder is linear at the input, so scaling is compensated by the weights; a scale of 0.15 changed the stream by 0.0007. The

TABLE IX
FUSION ON THE EXTERNAL CLINICAL CORPUS AS THE STRONGER STREAM IS WEAKENED. AREA UNDER THE ROC CURVE, MEAN OVER FIVE SEEDS IN EVERY ROW; GAINS ARE OVER THE BETTER SINGLE MODALITY, IN AREA UNDER THE ROC CURVE AND IN AVERAGE PRECISION (AP).

Condition	Vitals	Admission	Fused	Gain	Gain (AP)
Matched control	0.7068	0.8221	0.8241	$+0.0020$	-0.0008
Feature zeroed	0.6973	0.5947	0.6911	-0.0062	-0.0182
Titrated to parity	0.6992	0.6845	0.7282	$+\mathbf{0.0291}$	$+0.0121$

intervention used therefore mixes the feature with fixed-seed Gaussian noise at a ratio that preserves its variance, which removes information rather than rescaling it. A calibration on the admission stream alone, with two seeds, changed its performance monotonically from 0.6252 to 0.7859 across four mixing levels, and the level closest to the vital-sign stream was selected. Because the calibration uses only one single-modality arm and never the fused model, it does not select on the outcome. Because the current partitions contain $20\,\%$ less client data than the run reported above, the comparator was re-measured on them; its gain of 0.0020 confirms that the reduction does not affect the null result.

When the streams are balanced, a gain appears. With the two single-modality arms within 0.0147 of each other, the fused model attains 0.7282 ± 0.0191 against 0.6992 ± 0.0120 for the better single modality, a gain of 0.0291 ± 0.0211. The gain is positive on all five seeds, with a minimum of 0.0129 (paired t-test, $p = 0.019$; the signed-rank test gives $p = 0.0625$, its minimum attainable value at five seeds). The result exceeds the pre-registered threshold of 0.015 and is fourteen times the gain of the matched control. In average precision, the gain is 0.0121 ± 0.0215, with one negative seed, and is not resolved.

The scope of this result is limited in two respects. Balance was imposed by calibration, so the result shows that comparable stream strength is *sufficient* for a fusion gain on this corpus, but not that it is necessary. In addition, the gain is measured against a deliberately weakened stream: the balanced fused model remains well below the intact admission stream, so the result does not support fusing the two streams in the deployed configuration.

D. Necessity of the Components

Each component is removed in turn from the full system, with all other settings fixed, and the arms are paired by seed over three seeds (Table X). Of the four removals, only the removal of the privacy mechanism produces an interval that excludes zero.

1) Compression is justified by traffic, not by accuracy: Removing top-κ sparsification changes the area under the ROC curve by $+0.0012$, with an interval spanning zero, and increases uplink traffic by a factor of 5.0. Removing quantization changes it by $+0.0001$, with an interval of width 0.0011, the narrowest in the study, and increases uplink traffic by a factor of 1.60. Neither component affects predictive performance; together they reduce the payload eightfold at no measurable cost in discrimination.

TABLE X
COMPONENT ABLATION ON THE CLINICAL DEPLOYMENT. ΔAUROC IS PAIRED AGAINST THE FULL SYSTEM BY SEED OVER THREE SEEDS, REPORTED AS MEAN $\pm$ THE HALF-WIDTH OF A STUDENT-t INTERVAL; AN INTERVAL EXCLUDES ZERO WHEN THE HALF-WIDTH IS SMALLER THAN THE MEAN, AND THE ONE THAT DOES IS SET IN BOLD. MB IS THE TOTAL COMPRESSED PAYLOAD UPLOADED ACROSS TRAINING. C6 IS AN *inclusion* ARM: FOR IT A RISE, NOT A FALL, WOULD INDICATE A CONTRIBUTION.

Removed	AUROC	ΔAUROC	MB
— (full)	0.8541	—	252
C1 hierarchy	0.8484	-0.0058 ± 0.0088	50
C2 privacy	0.8307	$\mathbf{-0.0234 \pm 0.0205}$	252
C3 sparsification	0.8553	$+0.0012 \pm 0.0083$	1260
C4 quantization	0.8543	$+0.0001 \pm 0.0006$	403
+C6 ASWA	0.8542	$+0.0001 \pm 0.0009$	252

2) The edge tier does not cost accuracy: Removing the edge tier changes the area under the ROC curve by -0.0058 at three seeds, with an interval covering zero. With the six seeds specified in the pre-registration, the effect is resolved on both deployments: -0.0047 $[-0.0088, -0.0006]$ ($p = 0.032$) on the clinical corpus and -0.0052 $[-0.0074, -0.0030]$ ($p = 0.0018$) on the wearable corpus. Removing the tier therefore slightly reduces performance.

This result is interpreted as a bound on the cost of the hierarchy rather than as a benefit, because the flat arm differs in two respects. It uses a single edge with one edge round per global round, whereas the hierarchical arm uses five, so it changes both the aggregation schedule and the topology. The small advantage may be due to the additional aggregation steps rather than to the tier itself; separating the two would require a flat arm matched on aggregation steps. For the same reason, the uplink value of this arm is not comparable with the others. The comparison shows that the edge tier does not reduce predictive utility, so the communication and energy savings of Section V-A are not obtained at the expense of accuracy.

3) Removing the privacy mechanism does not recover utility: Training the same system without differential privacy *reduces* the area under the ROC curve by 0.0234 $[-0.0439, -0.0029]$ and increases the seed-to-seed spread in average precision from ± 0.014 to ± 0.070. Two of the three non-private seeds reach their best validation score in the first of ten rounds and degrade thereafter, whereas every private seed peaks between rounds six and nine. Clipping and calibrated noise therefore act as a regularizer for a compact head trained over frozen features, and without them the head overfits early.

This comparison is against a baseline without additional regularization beyond checkpoint selection, not against a non-private system tuned for generalization, which would typically use weight decay or early stopping. The result shows that removing the guarantee does not recover utility on this task; it does not show that differential privacy improves accuracy in general. It also shows that the accuracy cost of the guarantee, bounded at two percentage points in Table III, is not incurred at this operating point.

4) Loss-aware edge weighting: The aggregation rule ASWA, inherited from an earlier version of this framework,

TABLE XI
PATIENT-LEVEL MEMBERSHIP INFERENCE. MEAN ATTACK AUC OVER SEEDS WITH 95 % INTERVALS; 0.5 IS CHANCE. THE CONTROL IS NON-PRIVATE TRAINING OF THE SAME HEAD OVER THE SAME FROZEN ENCODERS.

Configuration	Attack	AUC	95 % CI
Control (non-private)	lik. ratio	0.6230	[0.598, 0.648]
Record-level DP	lik. ratio	0.5477	[0.488, 0.608]
Patient-level DP	threshold	0.5224	[0.518, 0.527]

weights each edge at the cloud by both its data volume and the mean local loss of its clients, instead of by data volume alone. It was evaluated as an inclusion arm. Adding it to the full system changes the area under the ROC curve by $+0.0001$, with an interval of width 0.0018 that covers zero, and leaves uplink traffic unchanged. Since this is an inclusion arm, an increase would indicate a contribution, and none is observed. The component is therefore not adopted.

E. Membership-Inference Audit

Table XI reports the empirical leakage of three configurations under both attacks of Section III-J. Each attack uses sixteen shadow models trained on independently sampled halves of the patient population, and each configuration is scored on the same 600 member and 600 non-member patients.

The positive control confirms that the audit can detect leakage. On non-private training of the same head over the same frozen encoders, the likelihood-ratio attack reaches 0.6230, compared with 0.5711 for loss thresholding. Per-patient calibration thus recovers 0.0519 of signal that a global threshold misses because of per-patient difficulty, as illustrated in Fig. 5.

The strength of the control does not depend on training length. Training for ten rounds instead of one changes the likelihood-ratio attack by -0.0047, and no attack by more than 0.006. A head of 560k parameters over frozen encoders therefore reaches its memorization capacity almost immediately, and a stronger control would require a different configuration rather than longer training.

On the private configurations, leakage is low. Record-level differential privacy yields 0.5477, with an interval that includes chance, and patient-level differential privacy yields 0.5224. The primary comparison, the paired difference between the two over five seeds on identical shadow splits, is not significant under any of the four attacks: $+0.0138$ $[-0.0034, +0.0310]$ for the loss threshold, $+0.0325$ $[-0.0280, +0.0929]$ for the offline likelihood ratio, $+0.0034$ for the online variant and -0.0211 for its pooled-variance form. All intervals span zero.

The *predicted* effect is therefore refuted. The hypothesis was that a record-level guarantee would be exploitable at the individual level, with an attack AUC approaching 0.7 against approximately 0.5 for a patient-level guarantee, and an effect of this size lies outside every interval obtained. A *small* effect is not excluded: at five seeds, the half-width for the strongest attack is ± 0.060, three of the four attacks point in the predicted direction, and the loss threshold reaches $p = 0.090$. The large predicted effect is thus absent, and an effect below approximately six points of AUC cannot be resolved at this sample size. The two guarantees cannot be considered equivalent on this evidence.

On the patient-level arm, the likelihood ratio performs *worse* than the loss threshold, by 0.0071. Since per-patient calibration helps only when difficulty-confounded signal is present, its benefit on the control and its absence on the patient-level arm indicate that the defense leaves little such signal. True-positive rates at a $1\,\%$ false-positive rate are low, at 0.029 on the control and 0.010 on the patient-level arm. At $0.1\,\%$, the rate is undefined, because 600 non-members are insufficient to resolve this false-positive rate.

The analytic argument of Section III-D therefore stands: a record-level budget bounds the wrong quantity, and its conversion to an individual-level budget is prohibitive, while the utility cost of enforcing the individual-level guarantee directly is small. The audit does not, however, show that the weaker guarantee is exploitable in practice on this data at this budget.

VI. DISCUSSION

A. What the Price of the Unit Depends On

The central finding of this paper is not that an individual-level guarantee has a cost, which was expected, but that the cost is not constant. At a single operating point, the guarantee reduces the area under the ROC curve by 0.0188. When the per-step budget and the number of units per step are held fixed, so that only the clipping geometry differs, the penalty is indistinguishable from zero at one record per individual and increases monotonically with the number of records, reaching 0.0866 in average precision at eight records and largely saturating before that point.

Two features of this measurement are important. First, the design contains its own control: at one record per individual, the two units coincide by construction, so the arms must agree, and they do (-0.0052, with an interval covering zero). The divergence at larger contributions can therefore be attributed to the unit. Second, the divergence is asymmetric. The individual-level arm remains flat across the whole range, whereas the record-level arm improves. The widening gap thus reflects the absence of improvement under the stricter unit rather than degradation, and an individual-level guarantee becomes relatively more expensive, not cheaper, as more records are collected per patient.

The external corpus supports the shape of this result but not its magnitude. On a second clinical corpus with real hospitals as clients, a median of 444 individuals per client and encoders pretrained on a verified disjoint population, the ordering is reproduced, from a negative gap at $k = 1$ to an increasing positive gap through $k = 8$. The registered saturation is also reproduced: from $k = 8$ to $k = 16$, the gap changes by -0.0030 $[-0.0316, +0.0255]$, so the cost approaches a bound in k. The magnitude, however, is approximately one sixth of the pre-registered forecast and is not resolved at three seeds. The shape of the law is therefore interpreted as structural and its scale as corpus-dependent, and the two corpora are not combined into a single estimate.

One possible explanation is that the two tasks differ in attainable precision by approximately a factor of two. Because a gap in average precision is bounded by the attainable precision, a cost that is a fixed *fraction* of attainable performance would produce a smaller absolute gap on the harder task with the same ordering in k. Under this explanation, the transferable quantity is the cost relative to attainable precision rather than the ceiling. A third corpus is required to distinguish this explanation from a corpus-specific ceiling.

The comparison between corpora is subject to a budget confound, which is bounded but not eliminated. The first corpus was originally trained at a higher budget, and additional noise reduces differences between arms. At the matched operating point, the gaps on the first corpus are not smaller, so at face value the budget explains none of the shortfall. The matched grid, however, fails its own $k = 1$ control, and under the conservative assumption that this offset applies at every k, the budget accounts for approximately half of the difference at $k = 8$. Under neither interpretation does it account for the factor of six, and under neither does the corpus term change sign.

The cost is also not an artifact of the clipping rule (Section V-B6). Clipping the summed contribution affects individuals with many records most strongly, but normalizing the clipping bound by the record count, which reduced the clipped fraction in every round from the third onward, did not improve utility, and the registered effect size is excluded by the one-sided upper bound. The cost is therefore a property of the individual-level unit. This conclusion rests on three seeds, one of which selected a first-round checkpoint, and applies only to the external corpus, since the first corpus has an almost constant number of records per individual.

B. What the Initialization Provides, and What the Privatized Subset Does Not

The proposed recipe differs from conventional private training in two respects: it starts from pretrained encoders and privatizes only the fusion head. Compared with full-model private training from a random initialization, it shows a large advantage that grows as the budget tightens. The three-arm decomposition shows that this advantage is due almost entirely to the initialization. Restricting the privatized subset contributes at most 0.0036 on the clinical corpus and 0.0013 on the wearable corpus, with intervals covering zero. Across the full budget range, the randomly initialized arm loses 0.1181 and 0.3716 on the two corpora, while the pretrained arm loses 0.0079 and 0.0157; on the wearable corpus, the randomly initialized arm falls to chance.

Two consequences follow. First, robustness to a tight budget depends on the initialization rather than on the fraction of the network that is privatized, so obtaining a legitimate pretrained initialization should take priority over selecting the privatized parameters. Second, head-only training cannot be justified by a reduction in noise. The noise scale under a given budget depends on the budget, the number of steps and the sampling rate, not on the number of parameters, so freezing most of the network does not reduce the noise per remaining parameter. Head-only training instead reduces communication, since only head parameters are transmitted, which yields the uplink reductions of $90.8\,\%$ and $96.1\,\%$ at no measurable cost in accuracy.

C. Which Components Are Necessary

In the component ablation, only the removal of the privacy mechanism produces an interval that excludes zero. The compression components do not affect accuracy: removing sparsification changes it by $+0.0012$ and increases uplink traffic fivefold, and removing quantization changes it by $+0.0001$ and increases the payload by three fifths. These well-powered null results support the use of both components for traffic reduction.

Removing the edge tier slightly reduces accuracy: the effect is not resolved at three seeds (-0.0058) but is resolved at the six pre-specified seeds, at -0.0047 $[-0.0088, -0.0006]$ on the clinical corpus and -0.0052 $[-0.0074, -0.0030]$ on the wearable corpus. This is not interpreted as evidence that hierarchical aggregation improves discrimination, because the flat arm performs one edge round per global round instead of five and therefore changes the aggregation schedule together with the topology. A flat arm matched on aggregation steps would be required to separate the two. The result establishes that the edge tier does not reduce predictive utility, so the uplink and energy savings are obtained without an accuracy penalty.

Removing the privacy mechanism *reduces* accuracy by 0.0234 and increases the seed-to-seed spread in average precision fivefold. Two of three non-private seeds peak in the first of ten rounds and then degrade, whereas all private seeds peak between rounds six and nine, which indicates that clipping and noise regularize a compact head over frozen features. This applies to a small head on frozen encoders and does not imply that differential privacy is without cost; the privacy-unit results quantify that cost under a different comparison. The tested inclusion arm was not adopted, as it produced no measurable effect.

D. Position Against Prior Art

The mechanisms used in this work are not new and are not claimed as contributions. User-level differential privacy, parameter-efficient private fine-tuning and differentially private quantile release are established. The open questions concern their combination and its cost. Hierarchical federated learning has been developed for energy and communication efficiency without a formal privacy guarantee, and private federated learning without a hierarchy, so the cost of including the edge tier in the privacy accounting has not been measured. Nor has the appropriate unit of the guarantee been examined for settings in which each individual contributes many records, which is the typical case in clinical and wearable sensing but uncommon in the benchmarks used to evaluate private federated learning.

The contributions are therefore measurements and the auditing practice under which they are reported, and several are

negative: the privatized subset contributes nothing once initialization is controlled, compression does not affect accuracy, an adaptive per-layer allocation of the budget performs worse than a uniform allocation because composition across groups is more costly, the tested inclusion arm provides no benefit, and the hypothesized difference in membership leakage between the two units is not of the predicted magnitude. Each is reported with the control or power analysis required to interpret it. In particular, the membership-inference null is accompanied by a positive control on a configuration known to memorize.

E. Practical Implications

Four recommendations follow from the measurements.

First, the unit of the guarantee should be chosen explicitly and its cost evaluated as a function of the number of records per individual. In settings where each individual contributes many records, a record-level budget does not bound the relevant quantity, and a post hoc group-privacy conversion is prohibitively expensive, whereas direct individual-level accounting costs noise rather than budget. The cost is predictable from the structure of the data and is bounded in the number of records per individual.

Second, a legitimate pretrained initialization should be obtained before other aspects of the private recipe are tuned. Under a tight budget, it has the largest effect of any decision measured here, and the disjointness condition required for an end-to-end guarantee has a measured and moderate cost.

Third, compression should be used. Both compression components have no measurable effect on accuracy and account for most of the uplink reduction, and the edge tier provides further communication and energy savings without an accuracy penalty.

Fourth, the benefit of fusion should be established per task rather than assumed from the availability of two sensors. The gain is 0.0370 in area under the ROC curve on the first clinical corpus, cannot be evaluated on the wearable corpus, whose better single modality already reaches 0.9976, and is absent on the external clinical corpus unless the two streams are balanced (Section V-C1).

F. Limitations

The privacy-unit results rest on two clinical corpora and cannot be tested on the third. In the wearable corpus, each client corresponds to one participant, so client-level, user-level and individual-level guarantees coincide and the divergence addressed by the mechanism cannot occur. The external clinical corpus reproduces the ordering and the saturation of the scaling law, but not its magnitude, which is approximately one sixth of that of the first corpus and unresolved at three seeds. The two corpora show that the law is not specific to one cohort, but they do not determine its constant, and no constant is reported.

The fusion gain is established on one corpus and, under imposed balance, on a second. The wearable corpus cannot test fusion, and on the external clinical corpus fusion yields no gain in its natural configuration. The titration shows that balance between the streams is sufficient for a gain in area under the ROC curve on that corpus, but not that it is necessary, and the gain is measured against a deliberately weakened stream. The average precision gain is not resolved. No unconditional fusion benefit is claimed.

Utilities on the two original corpora are obtained with the oracle encoder configuration. As stated in Section IV-B, the encoders for these results were pretrained on the pooled client training data, which does not provide a valid end-to-end guarantee. The deployable configuration costs 0.0322 in area under the ROC curve, so the absolute utilities on these corpora are upper bounds. Differences between arms, on which the claims rely, are unaffected because all arms share the same encoders. This limitation does *not* apply to the external clinical corpus, whose encoders were pretrained on $526{,}867$ windows verified to be disjoint from every client's training data.

The membership-inference null bounds only a large effect. The audit shows that the difference in leakage between record-level and individual-level guarantees is not of the hypothesized magnitude, but not that it is zero. At five paired seeds, the half-width for the strongest attack is ± 0.060 in area under the curve, three of the four attacks point in the predicted direction, and the strongest reaches $p = 0.090$. An effect below approximately six points of AUC would not be detected, and the two guarantees cannot be considered equivalent.

The audit uses sixteen shadow models. The per-individual in- and out-distributions are estimated from approximately eight observations each, which is why a pooled-variance variant is also reported. A larger shadow population would improve these estimates. In addition, 600 non-members are insufficient to resolve a $0.1\,\%$ false-positive rate, so the low-false-positive regime is characterized only at $1\,\%$.

One objective is inconclusive. On the wearable deployment, the accuracy cost under the budget is 1.967 pp with an interval of $[1.760, 2.173]$, which spans the 2 pp requirement. The seed count of fifteen was fixed in advance and was not increased, so the objective is reported as undetermined.

Several supporting measurements use few seeds. Each point of the privacy-budget sweep in Fig. 8 uses three seeds. This is sufficient for the initialization effect, which is much larger than the seed spread, but the near-zero difference between the two pretrained arms is bounded rather than measured, and the supported conclusion is that restricting the privatized subset has no detectable effect at this power. The aggregation-rule comparison that led to a component being excluded was based on one seed.

Only the network is simulated at packet level. Communication delay, delivery reliability and traffic volume are measured in a packet-level simulation, but the compute terms of the round latency (sensing, preprocessing, inference and aggregation) are modeled analytically and are identical across topologies. Comparisons between topologies are therefore valid, but the absolute round latency is not a measurement of a deployed system.

The interpretability evidence is associational. The attribution analysis shows that the model relies substantially on indicators of whether a measurement was taken, and that this reliance increases under the privacy mechanism. It does not

show that accuracy depends on these indicators, which would require removing them.

VII. Conclusion

This paper presented a hierarchical federated learning system for multimodal clinical and wearable sensing that enforces differential privacy at the level of the individual rather than the record, and quantified the cost of that choice. On two deployments, the composed budget, including the conformal calibration release, is 0.970 and 0.966. Uplink traffic is reduced by $90.8\,\%$ and $96.1\,\%$, energy per round by $72.3\,\%$ and $76.8\,\%$, and the 95th-percentile inference latency is 3.0 and 1.2 ms against a requirement of 100 ms. Because noise is added to gradients during training, the privacy mechanism adds no computation at inference.

Two findings are central. First, robustness to a tight privacy budget is determined by the initialization rather than by the size of the privatized subset. A pretrained initialization improves on a random one by 0.0187 without noise and by 0.1447 at the tightest budget, whereas restricting the privatized subset contributes at most 0.0036, with intervals covering zero, despite privatizing 3.3 times fewer parameters. The restriction is therefore justified by its communication savings only.

Second, the cost of an individual-level guarantee increases with the number of records each individual contributes. At a fixed per-step budget and number of units per step, the penalty is indistinguishable from zero at one record per individual and reaches 0.0866 in average precision at eight, saturating before that point. Because the two units coincide at one record per individual, this configuration serves as a built-in control. The individual-level arm remains flat while the record-level arm improves, so the cost arises from additional records that the stricter unit cannot exploit. Normalizing the clipping bound by the record count does not reduce this cost, which indicates that it is inherent to the unit rather than to the clipping rule. On a second clinical corpus of $132{,}736$ individuals across 187 hospitals, the ordering and the saturation transfer, whereas the magnitude is approximately one sixth of the pre-registered forecast and is not resolved at three seeds.

Several results are negative. Compression does not affect accuracy and is justified by traffic reduction alone. An adaptive per-layer allocation of the budget performs worse than a uniform allocation. The predicted difference in membership leakage between the two units is not observed under an attack whose positive control reaches 0.625. The benefit of multimodal fusion is conditional: it is 0.0370 in area under the ROC curve on the first clinical corpus, untestable on the wearable corpus, where the better single modality reaches 0.9976, and absent on the external clinical corpus. A titration on the external corpus shows that balance between the two streams is a sufficient condition: when the stronger stream is degraded to parity with the weaker, fusion yields a significant gain in area under the ROC curve over the better single modality, although not in average precision and not relative to the intact stronger stream.

Future work should establish whether the saturation level of the scaling law is a constant of the guarantee or a fixed fraction of the precision attainable on a task, which requires a third corpus. Establishing equivalence in leakage between the two units, rather than the absence of a large difference, requires a shadow population larger than sixteen and a non-member set large enough to resolve the low-false-positive regime.

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